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Control Question Selection for the Item Count Method

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Summary

Asking survey questions about sensitive topics — such as those involving criminal activity, sexual behavior, or racial prejudice — is an ongoing challenge. These questions suffer from higher-than-usual rates of misreporting, because respondents do not wish to admit to potentially embarrassing behaviors or attitudes. Survey researchers attempt to fix this issue by allowing respondents to hide their answer to sensitive questions in various ways. One way to accomplish this goal is with the item count method, a type of list experiment. The efficacy of the method depends on the list of "control" questions included alongside the sensitive question of interest. In particular, item count estimates are most efficient when the control questions are independent of the sensitive question. In this paper, I test whether that conditional independence can be met by using a list of control questions that are thematically unrelated to the sensitive question. I do so using a novel cross-national survey experiment where an item count question was administered using two control question lists: half of respondents were randomly assigned to see a list of thematically related control questions and the other half saw a list of thematically unrelated control questions. These data asked about whether respondents had illegally downloaded music, TV shows, or movies from the internet in the past year. I find that the choice of list does not results in substantively different estimates for the prevalence of internet piracy. Additionally, the thematically unrelated list is no better at meeting conditional independence than the thematically related list. These findings have important implications for questionnaire design for the item count method.

1 Introduction

Eliciting truthful answers to questions about sensitive subjects is a persistent methodological challenge in survey research. Sensitive topics include but are not limited to illegal drug use, sexual behavior, voting behavior, racial prejudice, and criminal activity. Respondents may see these topics as sensitive because they touch on a taboo subject. Alternatively, they could expect negative consequences from answering the question honestly (e.g., if they admit to committing a crime) or worry that their answer will be seen as socially unacceptable by the survey interviewer (Tourangeau & Yan, 2007). Survey researchers frequently wish to learn about sensitive behaviors and attitudes, including how common they are in the population and potentially how they correlate with other attitudes and behaviors.

However, survey questions about sensitive topics typically suffer from higher item nonresponse. More importantly, they are subject to high rates of misreporting — meaning respondents' purposely answer the sensitive question dishonestly. Generally, respondents overreport socially desirable behaviors like voting in elections and underreport socially undesirable behaviors like illegal drug use (Yan, 2021). As such, answers to sensitive survey questions are typically biased downward, meaning that population estimates derived from directly asking respondents about a sensitive behavior or attitude tend to be too low (Böckenholt, 2014).

Researchers have proposed a number of strategies to reduce bias and measurement error in sensitive survey questions. Most simply, self-administration of the sensitive question (as opposed to the question being asked by an interviewer) or asking the question in an open-ended format (Tourangeau & Yan, 2007). Theoretically, if respondents know their answer won't be seen by an interviewer, they will feel more comfortable answering honestly. Alternatively, researchers can "load" the introductory text of a question to make the sensitive topic seem more socially acceptable. For example, when asking respondents whether they voted in a recent election, one concern is that respondents will be embarrassed about not voting and answer dishonestly. Researchers therefore emphasize external factors that could prevent people from voting in the question text (American National Election Studies, 2021). Here is an example of such a question from the 2020 American National Election Study:

> "In talking to people about elections, we often find that a lot of people were not able to vote because they weren't registered, they were sick, or they just didn't have time. How about you—did you vote in the elections this November?"

However, these strategies are not entirely effective. Continuing the voting example, self-reported voter turnout is still subject to substantial overerporting even with tolerantly loaded introductions like the one above (Dahlgaard et al., 2019). Moreover, sensitive questions suffer from misreporting even in self-administered survey modes. As such, more comprehensive interventions have been developed. An early example of such a technique is the *randomized response method*, first proposed by Warner (1965). In this method, respondents use a randomization device that introduces a known chance of misclassifications to their response. This potential for misclassification hides their answer to the sensitive question from the survey interviewer (Kuha & Jackson, 2014).

Continuing the voting example above, respondents could be given a "spinner" that could either point to "I voted" or "I did not vote." The respondent would then spin the spinner and report only *whether the category it landed on was the category that applied to them.* Importantly, they would not reveal which category the spinner landed on. Not even the survey researchers can directly observe respondents' true answers to the sensitive question. But because the probability of the spinner landing on "I voted" or "I did not vote" is known, one can derive the maximum likelihood estimate (MLE) for the proportion of the population who voted (Warner, 1965).

Other variations of this method have been developed over time, as have methods for performing multivariate regression analysis on randomized response questions (Blair et al., 2015; Tourangeau & Yan, 2007). But while the randomized response method has been used successfully, it has important limitations. First and most obviously, respondents must perform the randomization themselves. Not only is the task relatively complicated, it is also a substantial departure from the standard rhythm of a survey interview. It can therefore introduce a "weirdness factor," where respondents become confused about why they are being asked to complete such an odd task (Kuha & Jackson, 2014). There is also evidence that the technique suffers from increased item nonresponse. More importantly, respondents can fail to carry out the spinner instructions properly or simply refuse to do so (Coutts & Jann, 2011; Holbrook & Krosnick, 2010b).

A newer technique that allows survey respondents to disguise their answers to sensitive survey questions is the *item count method* (ICM) (Miller, 1984). The technique takes the form of a survey list experiment (see Glynn (2013) for an overview of the item count method). Each respondent is shown a list of questions that can be answered either "Yes" or "No" (see Figure 1 for an example of how an item count question would be presented in a survey instrument). One of the items on this list is the sensitive question of substantive interest. The other items on the list are *control items.* Importantly, the control questions are not sensitive and are not of substantive interest. Respondents are not asked to give a"Yes' or "No' answer to each item individually. Instead, they only give their total number of "Yes" responses to the items on the list. They do not indicate the items to which they are responding "Yes" (Miller, 1984).

Like any survey experiment, the ICM works by randomly assigning half of respondent to the control group while the other half are assigned to the treatment group. The control group sees a list with *only the control questions.* The treatment group's list includes those same control questions *and* the sensitive question of substantive interest (Imai, 2011). Theoretically, respondents in the treatment group will feel comfortable giving an honest answer to the sensitive question because they must only report their total total number of "Yes" responses to the items on the list. Whether they answered "Yes" or "No" to the sensitive item itself is hidden from the interviewer and survey researchers (Kuha & Jackson, 2014). Figure 1 presents an example of an item count question from the substantive application I consider in this paper: illegally downloading music, TV shows, or movies from the internet.

The ICM has been shown to encourage survey respondents to answer sensitive questions truthfully, relative to asking the sensitive question directly (Li & Van den Noortgate, 2019; Rosenfeld et al., 2016; Tsuchiya et al., 2007). It is also preferable to the randomized response

method, primarily because respondents do not have to perform the randomization themselves (Imai, 2011). This change makes an item count question fit more naturally into a survey interview and reduces the burden placed on survey respondents substantially.

On the other hand, writing a list of control items in addition to the sensitive item of interest adds complexity to the survey design process. And more obviously, the item count method only solicits information about the sensitive question of interest from half of respondents (Kuha & Jackson, 2014). But despite these drawbacks, the ICM is an increasingly popular way to ask about sensitive behaviors and attitudes, and has been applied to study voter turnout, illegal drug use, and racial prejudice, and to conduct surveys in challenging research environments (Biemer & Brown, 2005; Blair et al., 2014; Holbrook & Krosnick, 2010a; Kuklinski et al., 1997). See Blair and Imai (2012) for a more detailed overview of how the ICM has been applied to substantive research questions.

I am now going to read you a list of five [six] things that people may do. Please listen to them and then tell me how many of them you have done in the last 12 months. Do not tell me which ones you have or have not done. Just tell me how many you have done at least once.

1. Visited a country outside [your country].
2. Gone to a football match.
3. Attended the wedding of a friend or a relative.
4. Gone to the dentist for a check-up.
5. Bought a new car.

[Item for treatment group only]
6. Downloaded music, TV shows or films from internet sites that may be illegal.

WRITE IN: ☐
(Don't know)

Figure 1: Example of an Item Count Method Question

There are two main quantities of interest from the item count method. First and most simply, estimates of the probability of the sensitive behavior or attitude occurring in the population. Second, regression-type estimates about how demographic characteristics or other explanatory variables are associated with the sensitive question (Imai, 2011; Kuha & Jackson, 2014). The most obvious problem in both cases is that respondents' answers to the sensitive question of interest are not observed. However, addressing this issue is relatively straightforward with the ICM.

Because the treatment group and control group should, in expectation, be identical except for the inclusion of the sensitive question in the item count list, one can use a moment-based estimator (the difference in mean reported count between the treatment group and control group) to estimate the sensitive probability of interest. The moment-based approach has also been extended to regression analysis (Coutts & Jann, 2011; Glynn, 2013; Holbrook & Krosnick, 2010a; Tsuchiya et al., 2007). However, moment-based item count estimation is potentially inefficient and is ill-suited for modeling a count response using regression (Kuha & Jackson, 2014).

As such, the ideal way to carry out item count estimation is to recognize that the method simply create missing categorical data. In other words, we wish to perform point estimation

and regression analysis, but some of the required data is not observed. Imai (2011) was the first to apply this missing data framework item count method. Under this approach, one treats respondents' answers to the sensitive question as missing data. We do not observe those answers directly. Instead, we only know (1) respondents' reported total of "Yes" responses to the entire list and (2) whether they were assigned to the control group or the treatment group.

When the problem is framed in this fashion, it becomes clear that estimation with ICM fits into the broader framework of regression and point estimation with missing data. As such, it is straightforward to perform point estimation using the expectation maximization (EM) algorithm (Dempster et al., 1977). Similarly, multivariate regression analysis can be carried out using either nonlinear least squares estimators or maximum likelihood (ML) estimators (Blair & Imai, 2012; Imai, 2011; Kuha & Jackson, 2014). This em-based approach was subsequently extended to allow for explicit variance estimation (something not typically provided when using the EM algorithm) and flexible models for the distribution of the control item total by Kuha and Jackson (2014).

Using the EM algorithm to perform item count point estimation and regression has been shown to work effectively (Imai, 2011; Kuha & Jackson, 2014). But like most survey experimental designs, item count estimates can be sensitive to questionnaire design choices (Ahlquist, 2018). Most importantly, one must design the list of control items that are paired with the sensitive question. There are some established best-practices for designing item count question lists (see Glynn (2013) for an overview). Most importantly, they should be designed to best meet the fundamental assumption of the method (see Section 2 for further discussion). But there remain unanswered questions about how to design the best possible item count lists.

I focus on one such question in this paper. If one can assume that respondents' answers to the control items are independent of their response to the sensitive question of interest (or are conditionally independent given a set of covariates), the ICM becomes more efficient (Kuha & Jackson, 2014). Clearly this is an ideal state, especially given that one only has data on the sensitive question from half of respondents. Therefore, to achieve the most precise ICM estimates possible, one should attempt to design item count lists so that the control items and sensitive item truly are independent (or conditionally independent). Unfortunately this cannot be tested *a priori*, and there is no previous work that identifies how to ensure independence (although Kuha and Jackson (2014) discuss the issue).

The goal is simple: design a set of control question that would be answered similarly by those who answer "Yes" and "No" to the sensitive question. The most obvious way to accomplish this goal is to make the control items thematically unrelated — meaning the subjects they address differ — to the sensitive item (Kuha & Jackson, 2014). Take as an example the sensitive topic of illegal cocaine use. Control questions that ask about alcohol consumption, smoking, or legal prescription drug use would likely correlate strongly with cocaine use. So instead, one should try to identify items that are less likely to be related to cocaine use, such as whether the respondent owns a dog or has flown on an airplane in the last year.

However, this question design approach has potential drawbacks. Respondents may be confused when presented with a list of seemingly unrelated items. They may not understand why the number of "Yes" responses to a list of disparate questions is of interest to the researchers, and the structure of the question breaks typical survey norms. Kuha and Jackson (2014) refer to this as the "weirdness factor" introduced by the item count method. The ICM is certainly less of a departure from survey norms than the randomized response method. But even still, the strangeness of the item count method can potentially alter how respondents would otherwise respond to the questions on the list. The most obvious way to reduce the ICM weirdness factor is by presenting a list of control items that *are* thematically related to the sensitive item. A list asking about cocaine use and alcohol consumption makes more sense than a list about cocaine use and dog ownership.

Obviously, there is a tension between these two design considerations. On one hand, we wish to achieve the most efficient estimates possible, in this case by making the control items independent of the sensitive item (Kuha & Jackson, 2014). On the other, we want to make the item count list seem as normal as possible. As such, using a thematically unrelated list of control items makes sense *only if* doing so allows one to assume that those control questions are conditionally independent of the sensitive question. But if the unrelated list *does not* allow one to assume conditional independence, using lists of control items that are thematically similar to the sensitive item would be preferable to reduce the ICM weirdness factor. It is also possible that these different control item lists could result in different estimates of the sensitive probability. We would not know which estimate is correct. But as a rule of thumb, the larger estimate is likely more accurate because sensitive questions tend to suffer from underreporting (Tourangeau & Yan, 2007).

As such, there are two main questions of interest. First, does using using a list of control items that is thematically dissimilar in content to the sensitive item allow us to assume conditional independence when that would not have been possible with a list of thematically similar control items? Second, do the related and unrelated control question lists result in significantly different estimates for the proportion of people in the population who would answer "Yes" to the sensitive question. And if the estimates do differ, which list produces the most accurate estimate?

I attempt to answer those questions in this paper using a novel survey from a cross-national survey conducted in the UK and Bulgaria that included an item count question on whether respondents had illegally downloaded music, TV shows, or movies from the internet in the last year. Critically, respondents within the survey were split into two groups. Group A was given an item count list with control items that were thematically unrelated to the sensitive question. Group B saw a list of thematically related control items. This design results in two separate item count list experiments. Below, I apply the EM algorithm to carry out item count estimation on these data, test whether the sensitive item estimates differ between lists A (unrelated) and B (related), and test whether we can assume independence between the control items and the sensitive question in either list. I also extend these analyses to the regression context to identify whether conditional independence can be achieved in either list with the inclusion of covariates.

The remainder of this paper proceeds as follows. In Section 2, I expand upon the item count method, introduce notation, and develop the idea of using EM for item count estimation. In Second 3, I discuss control question design choices for the ICM, focusing on achieving and testing conditional independence between the control questions and sensitive question. In Section 4, I introduce the survey dataset and item count question considered in this paper and discuss the methods of analysis used. Section 5 presents item count point estimates of internet piracy from my cross-national survey experiment. I also test for differences in the sensitive probabilities between lists A and B and test whether the control items and sensitive item are independent in either list. In Section 6, I provide the results of regression analyses on the internet piracy question using age as a predictor and introduce tests of conditional independence. Sections 7 and 8 contain additional analyses on variance estimation for the item count method, the selection of distributions for the control questions, and estimates of internet piracy separately for the UK and Bulgaria. Finally, Section 9 provides a discussion of my findings, the limitations of these analyses, and potential avenues for future research on the item count method.

2 The Item Count Method

2.1 Point estimation

When asking an item count question like the one presented in Figure 1, the primary goal is typically to estimate how frequently the sensitive behavior (in this case internet piracy) occurs in the population based on a sample of n survey respondents (Imai, 2011; Kuha & Jackson, 2014). The primary challenge is that one does not observe respondents' answers to the sensitive question directly. There are multiple ways to carry out point estimation for the item count method but I focus on two in this paper: moment-based estimation and the EM algorithm. Note that in this section, the notation I use to describe the item count method and EM algorithm was developed by Kuha and Jackson (2014). I use this notation for two reasons. First, the estimation approach I take follows the approach from that paper closely and the code used to perform the analyses I present here is based on the equations presented by Kuha and Jackson (2014). The purpose of this section is to describe how I derived the item count estimates presented in subsequent sections, not to develop a new approach to item count estimation. Second, it is useful to develop a common notation for the item count method. Doing so makes it easier to move between different papers on the subject.

An item count survey question consists of a list of J control questions and one sensitive question (Imai, 2011). We randomly assign n_0 respondents to the control group (who see a list of J control questions) and n_1 respondents to the treatment group who see a list of length $J+1$ that includes the sensitive question and the control questions (Kuha & Jackson, 2014). Note that $n_0 + n_1 = n$. We use the indicator $t_i, i = 1, ..., n$ to denote whether respondent i was randomly assigned to the control group ($t_i = 0$) or randomly assigned to the treatment group ($t_i = 1$).

We use Y_i to denote respondent i's answer to the sensitive question of interest. $Y_i = 1$ if respondent i answered "Yes" to that question and $Y_i = 0$ if they answered "No." This is our main quantity of interest. Similarly, let $Z_i = 0, ..., J$ denote the total number of control items to which respondent i answered "Yes." Critically, we observe neither Y_i nor Z_i separately, or indeed at all (Kuha & Jackson, 2014). Instead, the item count method provides respondents' reported "Yes" total to the entire list of questions with which they were presented. We denote this quantity as $S_i = Z_i + t_i Y_i$, with s_i denoting the value of S_i we observe for each respondent in our sample. For respondents in the control group, $t_i = 0$ so $S_i = Z_i$ with possible values $0, 1, ..., J$. But for treatment group respondents, $t_i = 1$ and $S_i = Z_i + Y_i$ with possible values $0, 1, ..., J + 1$ (Kuha & Jackson, 2014).

Now for even more notation preliminary notation from Kuha and Jackson (2014). We use $\mathbf{t} = (t_1, ..., t_n)'$ to denote the vector of treatment indicators. Similarly, $\mathbf{s}$ and $\mathbf{Y}$ represents the vectors of observed item count responses and answers to the sensitive question, respectively. Finally, for the later regression context, we take $\mathbf{X} = (x_1, ...x_n)'$, where each x_i is a vector of independent variables (including a constant) for respondent i (Kuha & Jackson, 2014).

Given the decomposition $S_i = Z_i + t_i Y_i$, we model S_i using two separate models. First, a model for the probability of answering "Yes" to the sensitive item of interest $p_y(y_i | x_{yi}; \beta)$, for $y_i = 0, 1$. Second, a model for the number of "Yes" responses to the control items:

$p_z(z_i|y_i, x_z, ; \psi)$, for $z_i = 0, 1, ..., J$ (Imai, 2011). Here, x_y is the vector of predictor variables used in the sensitive model and x_z is the vector predictor variables used in the control count model. Similarly, β is the vector of regression coefficients for the sensitive model and ψ is the same for the control count model. For simplicity, we can write $\theta = (\beta', \psi')'$.

We are primarily interested in the model for the sensitive item. After Kuha and Jackson (2014), I employ the binary logistic model shown below:

$$\pi_{yi} = p_y(y_i = 1|x_{yi}; \beta) = \frac{\exp(x_{yi}\beta)}{1 + \exp(x_{yi}\beta)} \tag{1}$$

This model can obviously be used to perform regression analysis on the sensitive item. But when the model has no predictor variables (aside from a constant) so $x_i = 1$, π_{yi} gives the overall probability of a "Yes" response to the sensitive question (Kuha & Jackson, 2014). For this approach to be valid, we must make the following assumptions (described in more detail by Kuha and Jackson (2014) and Imai (2011)):

1. *Randomization of the treatment*: The joint distribution of (Y_i, x_{yi}) is independent of t_i, so $p_y(y_i|x_{yi}; \beta) = p_y(y_i|x_{yi}, t_i; \beta)$.

2. *No liars*: Respondents' answers to the sensitive item Y_i are truthful.

3. *No design effect*: Respondents' total number of "Yes" responses to the control items Z_i are unaffected by whether the sensitive item was present in the list, so $p_z(z_i|y_i, x_z, ; \psi) = p_z(z_i|y_i, x_z, t_i; \psi)$.

4. *Nonresponse*: The probability of not answering the item count question is independent of (Y_i, Z_i) conditional on (x_{yi}, x_{zi}).

If these assumptions hold, Kuha and Jackson (2014) show that the model for the reported "Yes" total from the item count question S_i can be written as:

$$p_s(s_i|x_i, t_i; \theta) = \pi_{yi} p_z(s_i - t_i|y_i = 1, x_{zi}; \psi) + (1 - \pi_{yi}) p_z(s_i|y_i = 0, x_{zi}; \psi) \tag{2}$$

Importantly, there are two impossible values from this equation. First, $p_z(J + 1|y_i = 0, x_{zi}; \psi)$, which is impossible because it implies that a treatment group respondent selected every control item *and* the sensitive item (which they were not shown). Second, $p_z(-1|y_i = 1, x_{zi}; \psi)$, which would correspond to answering "Yes" to the sensitive question but having a reported item count total s_i of 0 (Kuha & Jackson, 2014). Because both outcomes are impossible, they occur with probability 0. That should be enforced when programming item count estimators.

The simplest way to carry out item count point estimation is by employing a moment-based estimator. In other words, the difference between the mean reported count s_i in the treatment group and mean reported count in the control group gives an estimate of the sensitive probability in the population (Kuha & Jackson, 2014):

$$\tilde{\pi}_y = \frac{1}{n_1} \sum_{i=1}^{n} t_i s_i - \frac{1}{n_0} \sum_{i=1}^{n} (1 - t_i) s_i \tag{3}$$

This estimate $\tilde{\pi}_y$ unbiased and (under certain conditions) is the maximum likelihood estimator for the population proportion (Kuha & Jackson, 2014). If desired, this saturated model can also be extended to acquire estimates of the control item totals conditional on responses to the sensitive item (Kuha & Jackson, 2014). This can be done using the equations below:

$$\tilde{p}_z(Z_i = j|Y_i = 0) = \big[\hat{p}_s(Z_i = j|Y_i = 1) - \tilde{\pi}_y \tilde{p}_z(Z_i = j-1|Y_i = 1)\big]/(1-\tilde{\pi}_y) \tag{4}$$

$$\tilde{p}_z(Z_i = j|Y_i = 1) = \big[\hat{p}_s(Z_i = j|Y_i = 0) - (1-\tilde{\pi}_y) \tilde{p}_z(Z_i = j|Y_i = 0)\big]/\tilde{\pi}_y \tag{5}$$

One calculates the probability of selecting each number of control items iteratively for $j = 0, ..., J$ starting with $\tilde{p}_z(Z_i = -1|Y_i = 1) = 0$ (see Glynn (2013) and Kuha and Jackson (2014) for a more detailed explanation).

But while this moment-based approach is straightforward, it does have drawbacks. First, moment-based estimates cannot accommodate models for Y_i or Z_i that are not saturated (Kuha & Jackson, 2014). More importantly, they are not ideal when one wishes to examine how the probability of answering "Yes" to the sensitive question is associated with respondents' characteristics or other attitudes. Simply using the moment-based estimation within demographic subgroups (e.g., calculating the difference in means separately for men and women) is often problematic (Imai, 2011). And while the model has been generalized to allow for linear regression (see Holbrook and Krosnick, 2010a), a linear probability model for Y_i is also unsatisfactory (Kuha & Jackson, 2014). Problems can also arise with the moment-based estimates for Z_i when assumption 3 is violated (see Section 3). As such, we turn to the EM algorithm.

2.2 Expectation maximization

As Imai (2011) first showed, one can rectify the issues mentioned above by treating item count analysis as a missing categorical data problem. When doing so, using the expectation maximization (EM) algorithm for point estimation and regression becomes the natural solution (Dempster et al., 1977). In this paper, I follow the EM estimation procedure developed by Kuha and Jackson (2014) and introduce a pseudo-dataset. The pseudo-dataset is simply two copies of the observed dataset with one stacked on top of the other. It therefore has $m = 2n$ rows. For each observation in the pseudo-dataset, row i and row $i + n$ have identical values on all observed variables, including s_i, t_i, and any covariates.

We then add two additional variables to the pseudo-dataset. First, y_i^*, which equals 1 for observations $i = 1, ..., n$ and 0 for observations $i = n + 1, ..., m$. Second, z_i^*, which equals $s_i - t_i$ for observations $i = 1, ..., n$ and s_i for observations $i = n + 1, ..., m$ (Kuha & Jackson, 2014). The result of these introductions is that rows $1, ..., n$ of the pseudo-dataset assume that every respondent gave a "Yes" to the sensitive question and the final $n + 1, ..., m$ rows assume that every respondent gave a "No" to the sensitive question.

Analogously to the models for y_i and z_i defined in the prior section, we define $p_y^* = p_y(y_i^*|x_{yi}; \beta)$ and $p_z^* = p_z(z_i^*|y_i, x_z, ; \psi)$ for the pseudo-dataset. Note that p_y^* equals π_{yi} (as

defined in Equation 1) for observations $i = 1, ..., n$. As such, p_y^* equals $1 - \pi_{yi}$ for observations $i = n + 1, ..., m$ (Kuha & Jackson, 2014). Respondents' answers to the sensitive questions Y_i are missing categorical data. We can therefore perform maximum likelihood estimation on the item count data using the expectation maximization (EM) algorithm (Dempster et al., 1977). If the sensitive answer Y_i and control count Z_i were observed separately, Kuha and Jackson (2014)show that we could write the *complete data log likelihood* in terms of the pseudo-data defined above as:

$$l(\theta; \mathbf{Y}, s) = \sum_{i=1}^{m} I(Y_i = y_i^*) \log p_{yi}^* + \sum_{i=1}^{m} I(Y_i = y_i^*) \log p_{zi}^* \tag{6}$$

In this equation, Y_i denotes the true response to the sensitive question for each of the n respondents, and $I(Y_i = y_i^*) = 1$ if $Y_i = y_i^*$. The pseudo-data includes two rows for each of the original n respondents, one where $y_i^* = 1$ and one where $y_i^* = 0$. Because Y_i is not observed, we do not know which of these rows is correct for each respondent. As such, the pseudo-data allows us to consider both possibilities when applying the EM algorithm (Kuha & Jackson, 2014). To carry out estimation of $\hat{\theta}$ (the regression parameters for the sensitive model and control model), one selects starting values $\theta_{(0)}$ and iterates between the following two steps until convergence (Dempster et al., 1977; Imai, 2011).

E-step

In this step, we calculate the expected value of Equation 6, conditional on the observed pseudo-dataset and the estimate of θ at iteration t, which we denote as $\theta_{(t)}$. Kuha and Jackson (2014) show that the expected log likelihood takes the following form:

$$Q(\theta|\theta_{(t)}) = Q(\beta|\theta_{(t)}) + Q(\psi|\theta_{(t)}) = \sum_{i=1}^{m} w_{i(t)}^* \log p_{yi}^* + \sum_{i=1}^{m} w_{i(t)}^* \log p_{zi}^* \tag{7}$$

This equation depends on the previously defined p_{zi}^* and p_{yi}^*, in addition to the weights $w_{i(t)}^*$. These denote the probability: $P(Y_i = p_{yi}^* | s_i, x_i, t_i; \theta_{(t)})$ (Kuha & Jackson, 2014). In simpler terms, we replace the indicators $I(Y_i = y_i^*)$ in Equation 6 (where we assumed the true value of Y_i was known) with these probabilities to allow for the possibility that $Y_i = 1$ and $Y_i = 0$ for each observation i. The $w_{i(t)}^*$ as defined by Kuha and Jackson (2014) are calculated using the parameter values at the current iteration $\theta_{(t)}$:

$$w_{i(t)}^* = \frac{p_{yi(t)}^* p_{zi(t)}^*}{p_{yi(t)}^* p_{zi(t)}^* + \big(1 - p_{yi(t)}^*\big) p_{zi'(t)}^*} \tag{8}$$

where $i' = i + n$ for $i = 1, ..., n$ and $i' = i - n$ for $i = n + 1, ..., m$.

M-step

Next, we simply maximize $Q(\theta|\theta_{(t)})$. Doing so provides updated parameter estimates $\theta_{(t+1)} = \big(\beta_{t+1}', \psi_{t+1}'\big)$. As shown in Equation 7, $Q(\theta|\theta_{(t)})$ decomposes into $Q(\beta|\theta_{(t)})$ and $Q(\psi|\theta_{(t)})$. As such, these terms can be maximized separately (Kuha & Jackson, 2014). Note that $Q(\beta|\theta_{(t)})$

is the weighted log likelihood for the model for Y_i and $Q(\psi|\theta_{(t)})$ is the same for the model for Z_i.

Therefore, we can acquire updated estimates by simply fitting two separate regression models to the pseudo-data, one for Y_i and one for Z_i. These models must include $w^*_{i(t)}$ as frequency weights. For $Q(\beta|\theta_{(t)})$, we fit a logistic regression model with y^*_i as the dependent variable and x_{yi} as the independent variables (Kuha & Jackson, 2014). For $Q(\psi|\theta_{(t)})$, we take z^*_i as the dependent variable and x_{zi} as the independent variables. The model for Z_i can take multiple forms. I discuss these options in Section 2.3.

Final EM estimates

Ideally, iterating between the E-step and M-step as described above will provide the maximum likelihood estimate $\hat{\theta}$ of θ once $Q(\theta|\theta_{(t)})$ converges (meaning the difference between successive Q terms is sufficiently small) (Dempster et al., 1977). But this is not guaranteed. The EM algorithm can diverge, so starting parameter values $\theta_{(0)}$ should be chosen carefully (Kuha & Jackson, 2014). In some cases, the algorithm should be run multiple times with different starting values to ensure it has not settled into a local maxima.

Provided the algorithm does converge, one can use the set of estimates $\hat{\theta}$ to examine regression coefficients as they would in a standard regression model without missing data (if explanatory variables are included) and derive an estimate for the population probability of responding "Yes" to the sensitive item of interest using Equation 1. Additionally, Kuha and Jackson (2014) show that the observed-data log-likelihood at the MLE is given by the pseudo-data as:

$$l(\hat{\theta}; s) = \sum_{i=1}^{n} \log\big(\pi_{yi} p^*_{zi} + (1 - \pi_{yi}) p^*_{z,i+n}\big) \tag{9}$$

with the probabilities evaluated at $\hat{\theta}$.

2.3 Modeling the control count

Choosing a model for the the sensitive item Y_i is straightforward. Y_i is binary, so we apply logistic regression as shown in Equation 1. But there are more options available when modeling the number of control items selected Z_i. Although this model and its coefficients are not of primary interest, it must be estimated, as the algorithm in the previous section involves $p_z(z_i|y_i, x_z, ; \psi)$ and probabilities p^*_{zi}. More importantly, misspecification of the model for Z_i can results in inaccurate estimates of the sensitive probability of interest (Kuha & Jackson, 2014).

When Imai (2011) first applied the EM algorithm to the item count method, the associated R software package (R Core Team, 2012) developed by Blair and Imai (2010) `list` allowed for two distributions for Z_i: binomial and beta-binomial. However, control count other models are possible. For example, one can model Z_i using a multinomial distribution and apply either ordinal or multinomial logistic regression. There are tradeoffs involved when choosing a distribution for Z_i. Using a simper model like the binomial requires one to estimate fewer parameters. However, Kuha and Jackson (2014) show that when the control count model

Z_i is correctly specified, EM estimators for the sensitive item of interest are approximately unbiased. More importantly, this also holds when the model is overparameterized. But a misspecified model for Z_i can result in dramatically biased EM estimates.

As such, they recommend that the default distributional assumption for Z_i should be multinomial to allow for maximum flexibility (Kuha & Jackson, 2014). Without prior knowledge, one should not by default consider only the special cases presented by the binomial distribution or the beta-binomial distribution (see Kuha and Jackson (2014) for a longer discussion of model selection for Z_i). In this paper, I follow the advice of Kuha and Jackson (2014) and consider a multinomial distribution with index 1 and probabilities $p_{zi} = p_z(z_i|y_i, x_{zi}; \psi)$ for $z = 0, ..., J$. When carrying out the M-step of the EM algorithm and modeling the relationship between Z_i and the possible explanatory variables, I employ a multinomial logistic regression model. I obtain the multinomial probabilities shown above by modeling $\pi_{zij} = P(Z_i = j|Y_i = y_i, x_{zi}; \psi)$ for all $j = 0, 1, ..., J$ (Kuha & Jackson, 2014).

Do to so, I introduce more notation from Kuha and Jackson (2014). We define δ_{ij} such that $\delta_{ij} = 1$ if $z_i = j$ and 0 otherwise. As such, $p_{zi} = \sum_{j=0}^{J} \delta_{ij}\pi_{zij}$. We also use $\mathbf{v}_i$ to denote the variables that make up the linear predictor of the model for Z_i. These can include y_i^* (when dependence between Y_i and Z_i is assumed), x_{zi}, and potential interactions between them. The multinomial probabilities can therefore be written:

$$\pi_{zij} = \frac{\exp(\mathbf{v}_i'\psi_j)}{1 + \sum_{l=1}^{J} \exp(\mathbf{v}_i'\psi_l)} \tag{10}$$

for $j = 1, ..., J$. And for $j = 0$, $\pi_{zi0} = 1 - \sum_{j=1}^{J} \pi_{zij}$, so $\psi = (\psi_1', ..., \psi_J')$.

Using the pseudo-data framework, we define π_{zij}^* as π_{zij} for observations $i = 1, ..., m$ in the pseudo-dataset. Similarly, $\delta_{ij}^* = 1$ if $z_i^* = j$ and 0 otherwise, for observations $i = 1, ..., m$ of the pseudo-dataset (Kuha & Jackson, 2014). Thus, to carry out the M-step, one would fit a multinomial logistic regression model to Z_i^* using the frequency weights $w_{i(t)}^*$ defined in Section 2.2. I take this approach for all EM models in this paper. In Section 7, I compare this multinomial approach to the more constrained binomial and beta-binomial distributions.

2.4 Variance estimation

I take two approaches to variance estimation in this paper. Primarily, I use the approach outlined by Kuha and Jackson (2014) to derive the observed information matrix for the item count data. The information matrix is not automatically provided by the EM algorithm. As such, this procedure relies on the fact that the function $Q(\theta|\theta_{(t)})$ from Equation 7 can be used to derive the information matrix (Oakes, 1999). Kuha and Jackson (2014) show that the information matrix for item count data can be written:

$$I(\theta_{(t)}) = -\left[\frac{\partial^2 Q(\theta|\theta_{(t)})}{\partial\theta\partial\theta'} + \frac{\partial^2 Q(\theta|\theta_{(t)})}{\partial\theta\partial\theta'_{(t)}}\right]_{\theta=\theta_{(t)}} \tag{11}$$

We can derive this matrix as follows, again using notation developed by Kuha and Jackson (2014). Note that the expression below are correct when modeling Z_i using a multinomial logistic regression model. As before, we denote $\theta = (\beta', \psi')'$ and take $Q(\theta|\theta_{(t)})$ from Equation 7.

We then let $\mathbf{X}_y = [x_{y1}, ..., x_{yn}]'$ and $\pi_y = (\pi_{y1}, ..., \pi_{yn})'$. Similarly, $\pi^*_{y(t)} = (\pi^*_{y1(t)}, ..., \pi^*_{yn(t)})'$, where $\pi^*_{yi(t)} = P(Y_i = 1|s_i, x_i, t_i; \theta_{(t)})$. Importantly, we see that $\pi^*_{yi(t)} = w^*_{i(t)}$ for $i = 1, ..., n$ (Kuha & Jackson, 2014). Finally, we define $\mathbf{V}_1$ as a matrix with rows v_i for observations $i = 1, ...n$ of the pseudo-dataset, where $y_i = 1$. Similarly, $\mathbf{V}_0$ has rows v_i for observations $i = n + 1, ..., m$ of the pseudo-dataset where $y_i = 0$. We also take $\mathbf{V} = [\mathbf{V}'_1, \mathbf{V}'_0]'$ (Kuha & Jackson, 2014).

Finally, we define $\mathbf{D}_1$ as a matrix with rows $\partial \log p^*_{zi}/\partial\psi'$ for $i = 1, ..., n$, and $\mathbf{D}_0$ as a matrix with rows $\partial \log p^*_{zi}/\partial\psi'$ for rows $i = n+1, ..., m$ (Kuha & Jackson, 2014). We similarly denote $\mathbf{D}_{1(t)}$ and $\mathbf{D}_{0(t)}$ using $\partial \log p^*_{zi(t)}/\partial\psi'_{(t)}$. Kuha and Jackson (2014) show that the formula for $\mathbf{D}_1$ — where $z_i = z^*_i = s_i - t_i$, $y_i = y^*_i = 1$, $\mathbf{v_i}$ are the rows of $\mathbf{V}_1$, and $\delta^*_{ij} = 1$ if $z^*_i = 1$ and 0 otherwise — can be written as:

$$\mathbf{D}_1 = \left[\text{diag}(\delta^*_{i1} - \pi^*_{zi1})\mathbf{V}_1 ... \text{diag}(\delta^*_{iJ} - \pi^*_{ziJ})\mathbf{V}_1\right] \tag{12}$$

for $i = 1, ..., n$. The formula for $\mathbf{D}_0$ is identical, except that the data are $z_i = z^*_i = s_i$, $y_i = y^*_i = 0$, and you use $\mathbf{V}_0$ instead of $\mathbf{V}_1$. $\mathbf{D}_{1(t)}$ and $\mathbf{D}_{0(t)}$ use these same equations but are evaluated at ψ_t. For the first term of (11), Kuha and Jackson (2014) show that we can write the derivatives with respect to β and ψ separately. The former is:

$$\frac{\partial^2 Q(\theta|\theta_{(t)})}{\partial\beta\partial\beta'} = -\mathbf{X}'_y \text{diag}\left[\pi_{yi}(1 - \pi_{yi})\right]\mathbf{X}_y \tag{13}$$

The latter is:

$$\begin{aligned}\frac{\partial^2 Q(\theta \mid \theta_{(t)})}{\partial\psi\,\partial\psi'} = &-\begin{bmatrix} \mathbf{V}'\,\text{diag}(w^*_{i(t)}\pi^*_{zi1})\,\mathbf{V} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \mathbf{V}'\,\text{diag}(w^*_{i(t)}\pi^*_{ziJ})\,\mathbf{V} \end{bmatrix} \\ &+\begin{bmatrix} \mathbf{V}'\,\text{diag}(w^*_{i(t)}\pi^*_{zi1}\pi^*_{zi1})\,\mathbf{V} & \cdots & \mathbf{V}'\,\text{diag}(w^*_{i(t)}\pi^*_{zi1}\pi^*_{ziJ})\,\mathbf{V} \\ \vdots & \ddots & \vdots \\ \mathbf{V}'\,\text{diag}(w^*_{i(t)}\pi^*_{ziJ}\pi^*_{zi1})\,\mathbf{V} & \cdots & \mathbf{V}'\,\text{diag}(w^*_{i(t)}\pi^*_{ziJ}\pi^*_{ziJ})\,\mathbf{V} \end{bmatrix}\end{aligned} \tag{14}$$

for $i = 1, ...m$ (Kuha & Jackson, 2014). This completes the first term of (11) because $\frac{\partial^2 Q(\theta|\theta_{(t)})}{\partial\beta\,\partial\psi'}$ and $\frac{\partial^2 Q(\theta|\theta_{(t)})}{\partial\psi\,\partial\beta'}$ and zero matrices. Finally, the second term of (11) is

$$\frac{\partial^2 Q(\theta \mid \theta_{(t)})}{\partial\beta\,\partial\beta'} = \left[\mathbf{X}'_y\left(\mathbf{D}_{1(t)} - \mathbf{D}_{0(t)}\right)'\right]\,\text{diag}\left[\pi^*_{y_i(t)}\left(1 - \pi^*_{y_i(t)}\right)\right]\,\left[\mathbf{X}_y\left(\mathbf{D}_{1(t)} - \mathbf{D}_{0(t)}\right)\right] \tag{15}$$

This term is symmetric when one evaluates $\mathbf{D}_1$ and $\mathbf{D}_0$ at $\theta = \theta_{(t)}$. That occurs when calculating the observed information matrix after the EM algorithm has converged (Kuha & Jackson, 2014).

With this information matrix, one can find the variance matrix for the estimates $\hat{\theta}$ by taking $\widehat{\text{var}}(\hat{\theta}) = I(\hat{\theta})^{-1}$ (Kuha & Jackson, 2014). Taking the square root of the diagonal of this matrix provides standard errors for the regression coefficients from the sensitive model $\hat{\beta}$

and from the control model $\hat{\psi}$. Subsequently, to derive a standard error for $\hat{\pi}_{yi}$ (my estimate for the unconditional probability of responding "Yes" to the sensitive question) I apply the delta method to the intercept coefficient from an intercept-only model for Y_i (see Oehlert, 1992). By default, I use these analytic standard errors for all EM algorithm models presented in this paper.

However, the most obvious alternative approach to producing variance estimates and standard errors for EM algorithm output is bootstrapping (see Horowitz, 2019). In this approach, one first resamples from the survey dataset of size n with replacement to construct a bootstrap sample, also of size n. The EM algorithm is then applied to the bootstrap sample and the coefficient estimates are saved. That process is repeated B times, where B is a suitably large number (100, 500, etc.). That process provides B distinct sets of coefficient estimates, one from each bootstrap sample. The variance of these bootstrap estimates can be used to approximate the variance of the true coefficient. Similarly, the square roots of those variances provide estimated standard errors for the regression coefficients.

While I primarily use the observed information matrix for variance estimation, I compare this approach to bootstrapping in Section 7. I first compare where standard errors from each approach agree and disagree. I also discuss the benefits and drawbacks of bootstrapping versus deriving the information matrix directly. In this paper, I implemented the above moment-based and EM estimation procedures in R (R Core Team, 2012). See the following link to download the code used in this paper: code download.

3 Control Question Selection

The previous section discussed how to carry out point estimation, regression, and variance estimation on item count survey questions using the EM algorithm. However, the effectiveness of the item count method (ICM) depends in large part on the design of the survey item itself (Kuha & Jackson, 2014). As mentioned in Section 1, the list experiment format can be strange for survey respondents. As such, it is important to predict how respondents will interpret item count list as a whole. Ideally, best-practices from the questionnaire design literature should be employed to design the most effective lists possible (Tourangeau et al., 2000). Much of the previous work on this topic has involved recommendations for designing item count lists that have the best chance at meeting the four major assumptions of the method articulated in Section 2.1 (see Glynn, 2013 and Blair and Imai, 2012 for detailed discussions of list design).

For example, to meet assumption 2 (no liars), one must avoid "ceiling effects." These occur when a treatment group respondent answers "Yes" to every control item. When that occurs, answering "Yes" to the sensitive item would necessarily reveal their answer, because they selected the maximum number of items available on the list (Kuha & Jackson, 2014). Other researchers have proposed statistical tests to detect violations of these assumptions. If assumption 3 (no design effect) is violated, some of the moment-based estimates for the control item count Z_i will be negative. But those estimates can also be negative due to sampling variation alone (that is another disadvantage of the moment-based approach). As such, Blair and Imai (2012) designed a test to identify whether negative moment-based estimates provide evidence of an assumption violation or could plausibly be caused by sampling variation.

In this paper, I do not focus on item count questions design choices meant to prevent violations of the four assumptions of Section 2.1. Instead, I assess approaches to questionnaire design that can affect the efficiency (and indeed accuracy) of ICM estimates. Beyond selecting a distribution for the control count Z_i (which here I model as multinomial), one must also choose which explanatory variables to include in the model for Z_i. The predictors for Z_i, denoted $\mathbf{v}_i$, always include x_{zi}. And often (but not always), $\mathbf{x}_{zi}$ includes the same variables as $\mathbf{x}_{yi}$. That is not strictly necessary, but it is the approach I take for the remainder of this paper. More importantly, $\mathbf{v}_i$ can also include respondents' answers to the sensitive question in the pseudo-dataset Y_i^* and possible interactions between Y_i^* and variables in x_{zi}.

Including Y_i^* as a predictor allows the number of "Yes" responses to the control items Z_i to depend on how respondents answered the sensitive question Y_i. In other words, it models the potential dependence between Z_i and Y_i. Importantly, Kuha and Jackson (2014) show that EM estimates for the item count method are most efficient when Z_i is independent (or at least conditionally independent) of Y_i, given the other predictors in $\mathbf{v}_i$. When conditional independence does not hold, both the sensitive model and control count model contain less information (Kuha & Jackson, 2014).

To see why, Kuha and Jackson (2014) note that second term of the information matrix $I(\beta)$ (see Equation 11) represents the missing information in the model due to Y_i not being directly observed. But the model for Z_i affects the amount of missing information present (Kuha & Jackson, 2014). When Z_i and Y_i are conditionally independent, only observations in

the treatment group contribute to that missing information (see Kuha and Jackson, 2014 for a more detailed discussion). In fact, a intercept-only EM algorithm item count model that uses a multinomial distribution as the model for Z_i produces identical estimates to the moment-based estimator in most cases. The only case in which the EM algorithm is more efficient than simple moment-based estimation is when we can assume conditional independence between Z_i and Y_i.

We cannot know whether conditional independence between Z_i and Y_i holds before conducting the item count survey. But ideally, one should design an item count question that maximizes the chances of Z_i and Y_i being independent. Kuha and Jackson (2014) argue that this goal should be achievable if the control items and the sensitive item are thematically unrelated. To continue the example from Section 1, if the sensitive question of interest is about illegal cocaine use, one would expect control questions about smoking, alcohol consumption, and legal drug use to correlate with "Yes" answers to the sensitive cocaine question (Mustaquim et al., 2021). But control questions about whether the respondent owns a dog, has flown on an airplane in the past year, or has seen a movie in theaters recently would have a better chance at being unrelated to the sensitive question. As such, to avoid introducing correlation between Z_i and Y_i, one should write control question that would be answered similarly by people who answer "Yes" and "No" to the sensitive question.

But that approach to list building is not without drawbacks. In general, an item count question departs from the standard survey question format respondents come to expect over the course of an interview (Tourangeau et al., 2000). Respondents might not understand why they've been presented with a list of seemingly-unrelated items, and might be confused why researchers care about the total number of those items that apply to them. We can call this the "weirdness factor" of the item count method after Kuha and Jackson (2014). Importantly, the strange question format can cause respondents to react to item count lists in unexpected ways. In general, survey researchers wish to reduce the weirdness factor to ensure respondents answer questions accurately. The most obvious way to accomplish this goal is by designing a list of control questions that make sense together and fit thematically with the sensitive item of interest. To continue the example above, a list about cocaine use, airplane flights, and movie theaters, and dogs would likely seem more out of place than a list about cocaine use, smoking, and alcohol consumption.

However, these two questionnaire design approaches are in almost perfect opposition to each other. By creating a list that fits together thematically, we risk introducing correlation between Y_i and Z_i. But if we design a list of thematically dissimilar items to achieve conditional independence, we likely increase the weirdness factor. There is no consensus in the item count method literature on whether control questions should be designed to be thematically similar or dissimilar from the sensitive question of interest. In this paper, my primary goal is to test which approach to list design is optimal by comparing two lists of item count control questions (one thematically similar to the sensitive question and one thematically dissimilar) used with the same sensitive item of interest.

I primarily employ two metrics to compare these control question lists. First, do the two lists produce significantly different estimates of the unconditional probability of responding

"Yes" to the sensitive question $\hat{\pi}_{yi}$? Ideally they should not, provided that respondents are assigned to the related and unrelated lists randomly. Clearly, if the lists do produce different estimates, we cannot know which is more accurate because Y_i is unobserved. But because responses to sensitive questions tend to be biased downward, a potential rule of thumb is that the higher of the two estimates is more accurate. Call the list of thematically unrelated question list A and the list of related questions list B. One can test whether they produce significantly different estimates by performing a Wald test on the difference between $\hat{\pi}_{yia}$ and $\hat{\pi}_{yib}$ (Wasserman, 2004). That takes the following form:

$$W = \frac{(\hat{\pi}_{yia} - \hat{\pi}_{yib})^2}{\widehat{\text{var}}(\hat{\pi}_{yia})^2 + \widehat{\text{var}}(\hat{\pi}_{yib})^2} \sim \chi^2_1 \tag{16}$$

Second, I assess whether conditional independence between Z_i and Y_i can be assumed for the thematically unrelated list (here called list A) but not the thematically related list (list B). If that is the case, it would provide evidence that using a list of thematically unrelated control items improves efficiency in item count estimation. To assess this claim, I perform a likelihood-ratio test between EM models with and without Y_i^* included as a predictor in the model for Z_i separately for each list (Wasserman, 2004). A statistically significant test would indicate that the conditional independence assumption does not hold. One can also examine the coefficients on Y_i^* in the control model directly. But because I model Z_i using multinomial logistic regression, the statistical significance (or lack thereof) of individual Y_i^* coefficients in models for each category of Z_i tell an incomplete story. The likelihood-ratio test takes the following form:

$$LR = -2\big[\ell(\theta_{\text{ind}}) - \ell(\theta_{\text{dep}})\big] \sim \chi^2_{\dim(\theta_{\text{dep}})-\dim(\theta_{\text{ind}})} \tag{17}$$

Where $\ell(\theta_{\text{ind}})$ and $\ell(\theta_{\text{dep}})$ come from Equation 9 for the models assuming independence and dependence, respectively. I now turn to the data to which I apply these tests.

4 Substantive Application: Pirating Music, TV Shows, or Movies

In this paper, I apply the methods outlined above to survey data from FIDUCIA, an EU-funded research project focused on public perceptions of new forms of criminal behavior, trust in justice, and the perceived legitimacy of justice institutions (Jackson et al., 2014). The project commissioned a cross-national survey of the United Kingdom (UK), Germany, Italy, Finland, Bulgaria, Lithuania and Turkey. I focus on data from the UK and Bulgaria waves that included my item count question of interest.

The UK survey was conducted between March 23rd and April 27th, 2015 in English among adults in England and Wales. Sampling was performed via telephone using Random Digit Dialing (RDD). The survey used quotas for region (North, Wales and South West, London, South East, and East and Midlands) and quotas for age, gender and working status within these regions. The core sample was based on landline telephones, supplemented by a list of mobile telephone numbers of individuals under age 35. Responses were collected using Computer Assisted Telephone Interviews (CATI), and a total of $1,004$ complete interviews were achieved.

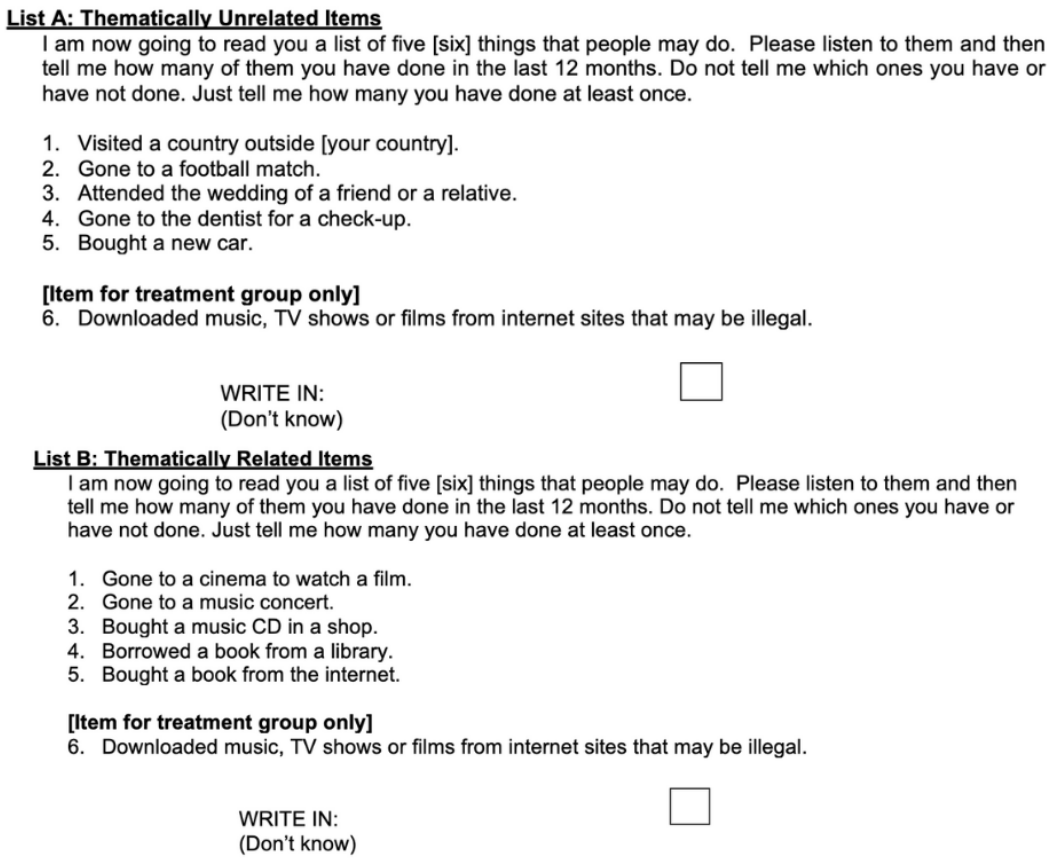
The Bulgaria survey was conducted between November 14th and December 4th, 2014 among individuals in Bulgaria over 16 years of age. A random sample was drawn using a two-stage stratified design. The survey used two stratification criteria: region (28 administrative districts and the 6 planning regions) and type of location (capital, regional city, town, or village). Within administrative areas, quotas were set based on age, sex, education and type of settlement. Responses were collected using face-to-face interviews at respondents' homes, and a total of $1,024$ complete interviews were achieved. See Appendix 2 for more information on the FIDUCIA survey.

The item count question in this survey focused on whether respondents had illegally downloaded music, TV shows or movies from the internet over the past 12 months. Researchers might wish to study how common that behavior is in the population, figure out what kinds of people are most likely to pirate digital media, or identify why people choose to engage in internet piracy. However, all of those goals require respondents to honestly answer whether they have downloaded digital media illegally. But that question is a sensitive one because it asks about a crime, and respondents may with to conceal that behavior. As such, the item count method can be employed.

Importantly, on top of the usual item count list experiment, there were two lists of control questions used for this item count question. The control items in list A were thematically dissimilar to the internet piracy question (e.g., going to a football match) and the questions in list B were thematically related to internet piracy (e.g., purchasing a music CD). Figure 2 below presents both lists. Half of the sample was randomly assigned to see list A and the other half to see list B. Within each list, the item count method proceeded as normal: half of respondents received the sensitive question in their list and half did not.

In effect, this design resulted in two item count questions, each with approximately $1,000$ survey respondents (pooling across the UK and Bulgaria samples). Both versions dealt with

the same sensitive topic (internet piracy) and differed only in the list of control items used. Assignment to list A (the thematically unrelated items) or list B (the thematically related items) was random. And within those lists, assignment to the treatment or control group was random. Therefore, we can assume that differences between the item count estimates from lists A and list B are due to the content of their control lists alone. These data are ideal to test whether thematically related and unrelated control items result in different estimates of the sensitive probability of interest, or whether we are able to assume conditional independence in the unrelated list but not the related list.



List A: Thematically Unrelated Items

I am now going to read you a list of five [six] things that people may do. Please listen to them and then tell me how many of them you have done in the last 12 months. Do not tell me which ones you have or have not done. Just tell me how many you have done at least once.

1. Visited a country outside [your country].
2. Gone to a football match.
3. Attended the wedding of a friend or a relative.
4. Gone to the dentist for a check-up.
5. Bought a new car.

[Item for treatment group only]

6. Downloaded music, TV shows or films from internet sites that may be illegal.

WRITE IN: ☐
(Don't know)

List B: Thematically Related Items

I am now going to read you a list of five [six] things that people may do. Please listen to them and then tell me how many of them you have done in the last 12 months. Do not tell me which ones you have or have not done. Just tell me how many you have done at least once.

1. Gone to a cinema to watch a film.
2. Gone to a music concert.
3. Bought a music CD in a shop.
4. Borrowed a book from a library.
5. Bought a book from the internet.

[Item for treatment group only]

6. Downloaded music, TV shows or films from internet sites that may be illegal.

WRITE IN: ☐
(Don't know)

Figure 2: Item Count Question Experiment from FIDUCIA Survey

As stated above, my primary goal is to identify whether one of these lists provides better estimates of the sensitive probability: in this case, the probability of illegally downloading music, TV, or movies from the internet. By better, I mean either more accurate, more efficient, or both. Therefore, I first compare the sensitive item estimates produced by the two lists, as described in Section 3. We obviously do not know the true value of Y_i. But we can assume that if one list of control questions provides a larger sensitive estimate, that estimate is the more accurate one. That assumption follows from the fact that people are far more likely to underreport sensitive behaviors than to overreport them (Tourangeau & Yan, 2007). I also test whether we can assume conditional independence between Z_i and Y_i in list A but not B. If that is the case, that would present evidence that using a thematically unrelated list of control items will provide more efficient item count estimates. In what follows, I carry out these tests on the full sample and separately for the UK and Bulgaria.

I also go beyond these primary tests of interest. First, I apply regression analysis to these data to assess how the probability of illegally downloading music, TV shows, and movies from the internet varies by age. Older individuals are less likely to frequently use the internet (Office for National Statistics, 2021). As such, my expectation is that there will be a strong negative relationship between age and the probability of illegally pirating media from the internet. I compare these models using AIC (defined as $-2\ell(\theta) + 2\dim(\theta)$). I also examine whether conditional independence can be assumed once age is included in the model.

Next, I compare the analytic standard errors described in Section 2.4 to bootstrap standard errors, and discuss their relative benefits and drawbacks. I then compare my estimates that assume Z_i follows a multinomial distribution to those from the `list` package in R that employs the binomial distribution and beta-binomial distribution for Z_i (R Core Team, 2012). You can see the code used in these analyses here: here.

5 Item Count Estimates of Internet Piracy

5.1 Moment-based estimates

Across both countries in the sample, the item count question was answered by 1,940 survey respondents. Of those, 991 were assigned to list A (the thematically unrelated list) and 950 were assigned to list B (the thematically related list). See Appendix 2 for more information on the sample. Table 1 presents the observed responses to the item count question about internet piracy in both lists, separately for the treatment group and control group. In the notation from Section 2 developed by Kuha and Jackson (2014) this table corresponds to s_i.

As expected, no respondents in the control group selected 6 items, as they were only presented with 5 options. Additionally, the observed counts do differ substantially between lists A and B. For example, respondents in list A were far less likely to select 0 options than respondents in list B. Those differences are also unsurprising given the differing control question lists. List A involves more general behaviors like going to the dentist, while list B is focused on media consumption habits that are likely rarer. As such, s_i contains more 0s in list B.

Table 1: Responses to the Item Count Question for the Treatment and Control Groups in Lists A and B

		Count							
List	**Group**	**0**	**1**	**2**	**3**	**4**	**5**	**6**	**Total**
List A	Control group	61	130	125	110	67	28	–	521
	Treatment group	66	99	92	100	80	24	9	470
List B	Control group	132	95	78	68	58	34	–	465
	Treatment group	127	97	82	63	66	33	17	485

Note: A count of 6 was only possible for the treatment group.

Table 2 presents the moment-based estimates for probability of illegally downloading music, TV shows, or movies $\tilde{\pi}_y$ and the probability of selecting between 0 and 5 of the control items Z_i, where $i = 0, ...5$, separately for lists A and B. Dependence between Z_i and Y_i must be assumed for these moment-based estimates, so Z_i is estimated separately for those who answered "Yes" and "No" to the sensitive question of interest. These are the simplest possible item count estimates and do not require use of the EM algorithm. They come from Equations 3, 4, and 5 in Section 2. The table also presents the difference in the $\tilde{\pi}_y$ estimates from list A and list B.

Based on list A, the estimated proportion of respondents who have downloaded music, TV shows, or movies from websites that are possibly illegal in the past year is 0.146. The estimate from list B is slightly higher at 0.180. As such, there does appear to be a slight difference between the two lists. As previously mentioned, the larger estimate would generally be considered more trustworthy because respondents tend to underreport sensitive behaviors. But the important question is whether a difference of -0.034 is enough to say that the lists

produce significantly different results. I examine that question in Section 5.2. Substantively, these results indicate that around 15% of people in the UK and Bulgaria have pirated digital media in the last year.

There is also evidence in Table 2 that the number of control questions to which respondents answered "Yes" (denoted Z_i) is indeed related to how they answered the sensitive question Y_i. For example, the probability of selecting 0 control questions is substantially higher for those who answered "No" to the sensitive question. That holds in both lists A and B. You will also notice a negative probability in list A. As mentioned in Section 3, that could either indicate a violation of the no design effect assumption or simply be caused by sampling error. These negative probabilities have boundary estimates of 0 (Kuha & Jackson, 2014).

Table 2: Moment-Based Estimates of the Probability of Illegally Downloading Music, TV Shows, or Movies from the Internet

			Control Count					
List	**Conditional on Y?**	$\hat{\pi}_y$	**0**	**1**	**2**	**3**	**4**	**5**
List A	$(Y = 0)$	0.146	0.164	0.274	0.211	0.179	0.131	0.040
	$(Y = 1)$		-0.160	0.107	0.410	0.399	0.113	0.132
List B	$(Y = 0)$	0.180	0.319	0.217	0.174	0.128	0.116	0.046
	$(Y = 1)$		0.123	0.146	0.139	0.230	0.167	0.195
Difference		-0.034						

These estimates were straightforward to acquire but are limited in important ways (Imai, 2011). Therefore in order to most accurately test whether the thematically related and unrelated lists produce significantly different item count estimates, I turn to estimation with the EM algorithm.

5.2 Comparing lists A and B

Table 3 presents the EM algorithm estimates for the item count question in list A and list B. Here, all models are fitted assuming that Z_i follows a multinomial distribution. But for each list, I ran the model twice: once assuming independence between Z_i and Y_i and once assuming dependence. This amounts to excluding or including Y_i^* as a predictor in the model for Z_i, respectively. The models that assume independence have only one set of control count estimates. But in the models that assume dependence there are two: one set for when $Y_i = 0$ and one for when $Y_i = 1$.

The models include no other predictors, so $\hat{\pi}_y$ represents the unconditional probability of responding "Yes" to the sensitive question about internet piracy. Across the four models, the estimated probability of downloading music, TV shows, or movies from illegal websites ranges from 0.180 to 0.250. Note that the sensitive probability for list B when we assume that Z_i depends upon Y_i (0.180) matches the moment-based estimate for list B. That is to be expected. The analogous EM estimate for list A does not match list A's moment-based estimate (0.205 vs. 0.146). The discrepancy is due to the negative probabilities in the

moment-based estimates for the control count (Kuha & Jackson, 2014).

For both lists A and B, the likelihood-ratio test between the models assuming dependence and independence rejects the null hypothesis that Z_i is independent of Y_i ($p = .019$ for list A and $p = 0.022$ for list B). Similarly, the AICs for these models indicate that the dependent models better fit the data. We can therefore conclude that Z_i does depend on Y_i in both lists. This dependence is also clear in the estimated control count probabilities, which vary substantially based on how respondents' answered the sensitive question. Respondents who answered "Yes" to the sensitive question are less likely to select 0 of the control questions and more likely to select all 5 of the control questions than respondents who answered "No" to the sensitive question. This occurs in lists A and B.

Additionally, when comparing the dependent models, the difference in $\hat{\pi}_y$ between lists A and B is only -0.051. The Wald test of that difference is not statistically significant ($p = 0.847$). The same holds when comparing the two independent models (difference between lists A and B = 0.026, $p = 0.494$). Based on these Wald tests, we can conclude that lists A and B do not produce significantly different estimates of the proportion of individuals who illegally download movies, TV shows, or music from the internet.

Table 3: EM Estimates of the Probability of Illegally Downloading Music, TV Shows, or Movies from the Internet

			Control Count							
List	**Conditional on Y?**	$\hat{\pi}_y$	**0**	**1**	**2**	**3**	**4**	**5**	**AIC**	***p*-value**
List A	Independent	0.199 (0.057)	0.141	0.238	0.220	0.214	0.135	0.052	3461.287	
	($Y = 0$)	0.205 (0.076)	0.161	0.269	0.209	0.182	0.137	0.042	3457.831	0.019
	($Y = 1$)		0.000	0.160	0.347	0.311	0.088	0.095		
List B	Independent	0.250 (0.047)	0.308	0.191	0.163	0.138	0.124	0.076	3374.470	
	($Y = 0$)	0.180 (0.110)	0.319	0.217	0.174	0.128	0.116	0.046	3371.332	0.022
	($Y = 1$)		0.122	0.146	0.139	0.230	0.167	0.195		
Difference	Independent	-0.051								0.494
	Dependent	0.026								0.847

Note: Standard errors in parentheses. The p-values in the rows for the list A and B models come from a likelihood-ratio test between the model assuming independence and the model assuming dependence. The p-values in the difference rows come from Wald tests of the differences between the list A and B sensitive estimates.

Based on these results, there is no evidence that using a list of control questions that are thematically related to the sensitive question of interest leads to more efficient EM estimates for the item count method. Even with the list of thematically unrelated questions, we are unable to assume that Z_i is independent of Y_i. Additionally, both lists produce substantively similar estimates of the sensitive probability $\hat{\pi}_y$.

Because there is no significant difference in the $\hat{\pi}_y$ estimates between lists A and B, it is theoretically possible to pool both lists together into a single dataset and estimate π_y using the full dataset. In practice, pooling two lists of control questions in a single model would be unnecessary, as only one list would be used per-sensitive question. But here, doing so provides extra statistical power and a more precise estimate of $\hat{\pi}_y$. This pooled dataset has

1,941 observations.

Table 4 presents estimates of π_y from the pooled dataset, using both moment-based and EM estimation. As expected based on the statistically insignificant Wald tests above, the estimates are substantively similar to those from the separate models for lists A and B. However, the standard errors for $\hat{\pi}_y$ are smaller. In the dependent model for example, the standard error is 0.052, compared to 0.076 and 0.110 for the list A and B dependent models, respectively.

Again, we can reject the null hypothesis that the number of control question to which respondents answer "Yes" is independent of how they answered the sensitive question. The p-value here ($p < 0.000$) is smaller than the p-values of the separate likelihood ratio tests for lists A and B, likely due to the larger sample size. As such, our best estimate of the proportion of people in the UK and Bulgaria who have illegally downloaded music, TV shows, or movies fron the internet comes from the pooled EM estimator. Here, that is 0.197. In other words, about 19.7% of the population of interest has engaged in internet piracy in the last year.

Table 4: Pooled Estimates of the Probability of Illegally Downloading Music, TV Shows, or Movies from the Internet

Model	**Conditional on Y?**	$\hat{\pi}_y$	**AIC**	p**-value**
Moment	—	0.152	—	—
EM	Independent	0.231 (0.037)	6834.2	—
	Dependent	0.197 (0.052)	6827.2	0.000

Note: Standard errors in parentheses. The p-value in this table comes from a likelihood-ratio test between the model assuming independence and the model assuming dependence.

Performing pooled estimation is not as simple as combining the list A and B datasets and applying the same R function used to perform EM estimation on an item count question with a single list of control questions. That's because the estimated probabilities for the control question counts are still expected to vary between the two lists even if the sensitive probability does not. It is theoretically possible that the control count probabilities between two lists do not differ significantly. But it is incredibly unlikely when, as in this application, the two lists differ in content. As such, I constructed a new R function that estimates the sensitive probability using the full dataset but fits separate multinomial logistic regression models to estimate the control count probabilities for lists A and B (R Core Team, 2012). The control probabilities associated with the sensitive estimates in Table 4 are shown below in Table 5.

Substantively, these estimates are similar to the control count probabilities from the separate models for lists A and B shown in Table 3. That holds for both the moment-based and EM estimators. This is entirely expected, as the control models are still estimated separately for each list. But the probabilities do differ slightly in the pooled model. That is because the control count models incorporate the weights $w^*_{i(t)}$ (see Equation 8). But $w^*_{i(t)}$ depends in part on our estimate for the sensitive probability. Therefore, the fact the pooled

estimate of $\hat{\pi}_y$ differs somewhat from the sensitive estimates from lists A and B means that the sensitive probabilities also differ.

Table 5: Pooled Moment-Based and EM Estimates of Control Count Probabilities by List

Model	**Conditional on Y?**	**0**	**1**	**2**	**3**	**4**	**5**
List A							
Moment	$Y = 0$	0.166	0.276	0.212	0.180	0.132	0.041
	$Y = 1$	–0.154	0.102	0.393	0.382	0.108	0.126
EM	Independent	0.143	0.239	0.220	0.215	0.133	0.051
	$Y = 0$	0.160	0.267	0.210	0.183	0.137	0.043
	$Y = 1$	0.000	0.156	0.347	0.313	0.087	0.098
List B							
Moment	$Y = 0$	0.309	0.210	0.168	0.124	0.112	0.045
	$Y = 1$	0.145	0.173	0.164	0.272	0.197	0.231
EM	Independent	0.306	0.191	0.163	0.138	0.125	0.077
	$Y = 0$	0.324	0.217	0.172	0.126	0.115	0.046
	$Y = 1$	0.128	0.155	0.148	0.227	0.161	0.180

Overall, what conclusions can we draw about our primary questions of interest? First, both lists A and B provide similar estimates of the probability that someone has illegally downloaded music, TV shows, or movies in the past year. Based on these data, the estimates are not significantly different. Second, we cannot assume that responses to the control questions Z_i are independent of responses to the sensitive question Y_i for either list. Therefore, it does not appear that using a list of control questions that are thematically unrelated to the sensitive question of interest provides a tangible benefit for item count estimation. Estimates using that list are neither more accurate nor more efficient. As such, for the substantive application in this paper, the pooled results from Tables 4 and 5 provide the best estimates for the frequency of internet piracy. More broadly, because the unrelated list provides no significant benefit, these data indicate that control items that are thematically similar to the sensitive question of interest should be employed when asking item count questions to reduce the "weirdness factor" that comes from using unrelated items.

6 Item Count Regression

6.1 Estimating how age relates to internet piracy

A major benefit of using the EM algorithm for item count estimations is the ease with which regression analysis can be performed. The EM estimates derived in the previous section come from intercept-only regression models (plus Y^* in some cases). It is therefore trivial to add additional predictors and examine how a variable of interest is related to the sensitive question. Although respondents' answers to the sensitive item are unobserved, we can still estimate how internet piracy relates to respondents' demographic characteristics and other attitudes. Importantly, it is possible that Z_i is conditionally independent of Y_i, given the introduction of particular covariates. If we can achieve conditional independence using the list of unrelated control items (list A) but not the list of thematically related items (list B), that would provide evidence that unrelated items do have the potential to produce more efficient item count estimates.

Here, I introduce age as an independent variable in both the model for the sensitive item Y_i and the control count model for Z_i. The substantive hypothesis is obvious: there should be a strong negative relationship between age and the probability of pirating media from the internet. Most simply, that is because older individuals are less likely to frequently use the internet (Office for National Statistics, 2021). Therefore, they are less likely to be interested in downloading music, movies, or TV shows at all. One would also expect older individuals to be less digitally skilled than younger individuals, at least on average. And there is past evidence that knowledge of and comfort with computer use is related to digital piracy (Eisend, 2019). Additionally, age is likely related to many of the items included in the control item lists: visiting another country, buying a new car, borrowing a book from the library, going to a music concert, etc. It is possible that the number of control questions selected Z_i will be independent of answers to the sensitive question Y_i once age is added to the model.

I include age in the EM model as a categorical variable with three categories: $16-34$, $35-64$, and $65+$. In all models, I treat $16-34$ as the reference category. I begin by presenting the results separately for lists A and B, both times assuming that Z_i is dependent on Y_i by including Y^* as a predictor. I do so because in Section 5, I found that the null hypothesis of independence between Z_i and Y_i was rejected for both lists. In this case, we are primarily interested in the regression coefficients for the sensitive and control models, not the estimated probabilities of engaging in piracy. Table 6 presents the regression coefficients for these models.

Starting with the model for the sensitive item Y_i, there is clear evidence that age is related to internet piracy. For example, in list A, the odds of illegally downloading music, TV, or movies are 96% lower for those who are 65 or older, compared to those who are younger than 35. That relationship is statistically significant at the 5% level. Although the coefficient for being aged $35-64$ goes in the same direction (-2.590), it is not statistically significant. We see a similar pattern in list B. The $35-64$ coefficient is negative but not statistically significant. The $65+$ coefficient is much larger (-16.505), but in this case, it has an enormous standard error (842). Generally in logistic regression, one would associate this behavior with

perfect separation, where the binary 65+ variable perfectly predicts the binary Y_i outcome (Rainey, 2016). In other words, no one aged 65 or older in list B answered "Yes" to the internet piracy question. I return to this issue shortly.

Table 6: Regression Results for List A and B Assuming Dependence between Z_i and Y_i

Model for Sensitive Item Y	
Predictor	**Estimate (SE)**
List A	
Intercept	0.366 (0.600)
Aged 35 to 64	–2.590 (1.710)
Aged 65 or over	–3.428 (1.443)
List B	
Intercept	0.120 (0.768)
Aged 35 to 64	–2.114 (1.399)
Aged 65 or over	–16.505 (842.090)

Model for Control Count Z					
Predictor	$Z = 1$	$Z = 2$	$Z = 3$	$Z = 4$	$Z = 5$
List A					
Intercept	1.590 (0.395)	1.186 (0.597)	–0.095 (1.548)	0.114 (1.096)	–2.620 (2.273)
Aged 35 to 64	–0.721 (0.432)	–0.396 (0.644)	0.860 (1.673)	0.301 (1.218)	1.639 (2.553)
Aged 65 or over	–1.860 (0.434)	–1.852 (0.635)	–0.532 (1.602)	–1.424 (1.184)	–0.015 (2.216)
Y^*	7.684 (179.604)	8.658 (179.610)	9.725 (179.594)	8.936 (179.603)	10.853 (179.598)
List B					
Intercept	0.649 (0.386)	0.299 (0.598)	0.176 (0.825)	–0.140 (0.742)	–1.576 (1.231)
Aged 35 to 64	–0.907 (0.407)	–0.747 (0.569)	–0.872 (0.736)	–0.568 (0.700)	0.100 (1.331)
Aged 65 or over	–1.659 (0.429)	–1.605 (0.633)	–2.076 (0.868)	–1.642 (0.785)	–0.900 (1.279)
Y^*	–0.149 (0.767)	0.178 (1.151)	0.524 (1.290)	0.419 (1.175)	1.711 (1.430)
Log-likelihood (List A)		–1671.116			
Log-likelihood (List B)		–1615.225			

Note: Standard errors in parentheses.

We also see that there is a relationship between age and answers to the control question. The 65+ coefficient is always negative across both models, but it is not always statistically significant at the 5% level. In both lists, respondents 65 and older are significantly less likely to have selected 1 item instead of 0 items, than are respondents under age 35. The same holds for selecting 2 items instead of 0 items, and for selecting 3 items instead of 0 in list B.

The $35-64$ coefficient is also mostly negative, but only shows a significant difference between that group and those younger than 35 in list B when comparing selecting 1 item to 0 items. Therefore, we can conclude that individuals over age 65 are less likely to commit internet piracy than those under age 35. We also have some evidence that individuals over 65 are less likely to do the things mentioned in both control lists (such as go to a concert or purchase a CD for list B, or go to a football match or buy a new car for list A).

We can also examine the Y^* coefficients in the model for Z_i. These quantify whether the number of control questions selected depends on how respondents' answered the sensitive question. In list B, we see that the coefficients are not statistically significant for any category and tend to be relatively small. In list A, the coefficients are much larger (they range from 7.68 to 10.85) but have enormous standard errors around 179. Again, that behavior is indicative of some kind of separation (an issue I address shortly).

But from the list B results, it is tempting to conclude that in introducing age as a covariate makes responses to Z_i conditionally independent of Y_i, thereby fixing the issue identified in Section 5. However, further examination is required, as Y^* may still contribute to the overall fit of model even if the individual coefficients are not statistically significant. Indeed, even in the intercept only model, many of the Y^* coefficients are not themselves statistically significant. But as seen in Table 3, the model that assumes dependence between Y_i and Z_i was clearly preferred.

6.2 Testing conditional independence

For the efficiency gains described in Section 3, one must assume that the number of control items selected Z_i is independent of the sensitive item Y_i or that Z_i is *conditionally* independent of Y_i given some covariates (Kuha & Jackson, 2014). To test whether adding age as a predictor has allowed us to achieve conditional independence in either list, we can use the same approach taken for the intercept-only models in Section 5. First, I fit the EM algorithm model twice to each list, once including Y^* as a predictor in the model for Z_i and once without including Y^*. I then examine the AICs of these models and perform a likelihood ratio test comparing the dependent and independent models for each list. This output is shown in Table 7.

In list A, the AIC for the dependent model ($3,388.232$) is nearly identical to the AIC for the independent model ($3,388.323$). Similarly, the likelihood-ratio test between the dependent and independent model fails to reject the null hypothesis of independence at the 5% level. It is, however, statistically significant at the 10% level level ($p = 0.073$). In list B, the independent model ($3,272.450$) actually fits the data better than the dependent model ($3,276.469$) based on the AIC. And in this case, the likelihood-ratio test is clearly not statistically significant ($p = 0.304$).

Table 7: Tests of Conditional Independence in EM Regression Models

List	Model	AIC	Test Statistic	p-value
A	Dependent	3388.232	10.091	0.073
	Independent	3388.323		
B	Dependent	3276.450	6.019	0.304
	Independent	3272.469		

Note: Test statistics and p-values are from a likelihood-ratio test between the independent model and dependent model.

Therefore, we can tentatively conclude that by including age as a predictor in the models for Y_i and Z_i, we can safely assume that Y_i and Z_i are independent of each other. As such, when age is included in the model, we can exclude Y^* as a predictor and achieve more efficient item count estimates. These results also provide additional evidence that the choice of control item list does not affect our results. Indeed, the likelihood ratio test for list A (the list of thematically unrelated items) is still marginally statistically significant, but it is clearly insignificant for list B (the related items). As such, we can again conclude that using a list of thematically unrelated control items does not provide noticeable efficiency gains.

Given these findings, Table 8 presents regression coefficinets from simplified EM regression models that exclude Y^* as a predictor variable. Broadly, the substantive conclusions from the regression models assuming independence between Z_i and Y_i and similar to the earlier models where independence was not assumed. However, as expected, the coefficients are estimated more precisely. In list A, there is a statistically significant negative relationship between being 65 or older and internet piracy. The $35-64$ coefficient is not statistically significant at the 5% level in list A, but now is significant in list B. In the model for the control count Z_i, the 65+ coefficients are again consistently negative, but they are now statistically significant at the 5% level for every category in both lists A and B. Additionally, the $35-64$ coefficients are now consistently negative. The are statistically significant for all categories in list B, but significant for only some list A categories.

Table 8: Regression Results for List A and B Assuming Independence between Z_i and Y_i

Model for Sensitive Item Y	
Predictor	**Estimate (SE)**
List A	
Intercept	–0.312 (0.560)
Aged 35 to 64	–0.837 (0.760)
Aged 65 or over	–2.766 (1.163)
List B	
Intercept	0.352 (0.510)
Aged 35 to 64	–1.510 (0.651)
Aged 65 or over	–15.358 (634.905)

Model for Control Count Z					
Predictor	$Z = 1$	$Z = 2$	$Z = 3$	$Z = 4$	$Z = 5$
List A					
Intercept	1.800 (0.372)	1.851 (0.372)	1.503 (0.387)	1.120 (0.415)	0.222 (0.498)
Aged 35 to 64	–0.994 (0.416)	–1.166 (0.419)	–0.710 (0.428)	–0.784 (0.474)	–0.940 (0.572)
Aged 65 or over	–2.088 (0.417)	–2.507 (0.426)	–2.057 (0.436)	–2.334 (0.489)	–2.543 (0.637)
List B					
Intercept	0.453 (0.290)	0.207 (0.274)	0.334 (0.277)	–0.124 (0.313)	–0.479 (0.339)
Aged 35 to 64	–0.842 (0.334)	–0.761 (0.317)	–1.067 (0.323)	–0.655 (0.357)	–0.804 (0.399)
Aged 65 or over	–1.462 (0.345)	–1.513 (0.344)	–2.234 (0.385)	–1.658 (0.403)	–1.997 (0.485)
Log-likelihood (List A)		–1676.161			
Log-likelihood (List B)		–1618.235			

Note: Standard errors in parentheses.

But again, we see that the 65+ coefficient is large (-15.358) and its standard error is enormous (634.905) in list B. This — and the standard errors for Y^* in the dependent model for list A shown in Table 6 — indicates perfect separation issues. In other words, it appears that in list B, no respondents over age 65 answered "Yes" to the sensitive question, and that a similar issue occurs to for responses to the control questions in list A for those who answered "Yes" to the sensitive question. In a standard logistic regression application, checking that the unusually large standard errors are caused by perfect separation (instead of a coding error, for example) is as simple as performing a crosstabulation between age and Y_i (Rainey, 2016). However, we have an added complication because Y^* is unobserved. I address that issue in the following section.

7 Standard Errors & Control Distributions

7.1 Calculating standard errors

The standard errors for the regression coefficients above come from the information matrix derived in Section 2.4. Similarly, the standard errors for $\hat{\pi}_y$ presented in Section 5 come from applying the delta method to the standard errors for the intercept in an intercept-only EM model for Y_i (Oehlert, 1992). That is the approach I take for the rest of the paper. But in Tables 6 and 8, we observed strange behavior from these standard errors that one would usually attribute to perfect separation. For example, because 65+ perfectly predicts internet piracy in list B, the information matrix is nearly singular and the standard error explodes. But because because Y^* is unobserved, it is harder than usual to confirm that the problem is indeed driven by separation.

Here, I resolve that difficulty by deriving standard errors for the EM regression coefficients using a different approach: nonparametric bootstrapping. I then compare the standard errors calculated analytically (as above) to those derived from bootstrapping. If the large Y^* and 65+ SEs are indeed caused by separation, the bootstrap SEs for those coefficients should also be abnormally large. More broadly, this comparison allows us to discuss which approach to variance estimation is preferable when implementing the EM algorithm for item count estimation.

I first focus on the intercept-only models presented in Table 3, assuming dependence between Z_i and Y_i. Because the Y^* coefficients exhibited separation-like behavior, I compare the bootstrap and analytic standard errors for those coefficients. But the results below should generalize to other model specifications. I first took 500 bootstrap samples from the list A dataset and applied the EM algorithm to each one. This resulted in 500 sets of regression coefficients for the sensitive and control regression models. I used these to approximate the sampling distributions for these coefficients, and took their standard deviations to derive standard errors for each coefficient. I then repeated this process for the list B dataset. Table 9 presents the standard errors for each coefficient separately for list A and B, using both the observed information matrix and bootstrapping.

For the intercepts, the bootstrap and analytic standard errors match closely in most cases. Where there is a difference, the magnitude is relatively small. For example, for the intercept for the $Z = 5$ multinomial control category in list B, the analytic SE is 0.39, while the bootstrap SE is 0.57. The direction of the difference is also not consistent. For example, the analytic SE for the sensitive intercept in list B (0.74) is larger than the corresponding bootstrap SE (0.54). There are, however, larger differences between the analytic and bootstrap SEs for the Y^* coefficients. In list A, the analytic Y^* SEs are enormous (around 36), which is clear evidence of a separation issue. The corresponding bootstrap SEs are also large (ranging between 2.3 and 3.4), but less so. This difference is likely not an issue. Instead, the analytic and bootstrap standard errors both pick up the instability in the $Y*$ coefficients, but in different ways. The list B results are similar. There, the bootstrap SEs (ranging between 2.3 and 3.1 are larger than the analytic SEs (1.1 and 1.4). But again, both indicate that the effect of Y^* on Z_i is weakly identified.

Table 9: Comparison of Analytic and Bootstrap Standard Errors for Intercept-Only Model

Coefficient	**List A**		**List B**	
	Analytic SE	Bootstrap SE	Analytic SE	Bootstrap SE
Model for sensitive item Y				
Intercept	0.463	0.446	0.744	0.542
Model for control count Z				
Intercept (Z = 1)	0.129	0.127	0.153	0.161
Intercept (Z = 2)	0.196	0.193	0.212	0.222
Intercept (Z = 3)	0.216	0.218	0.273	0.394
Intercept (Z = 4)	0.209	0.219	0.246	0.256
Intercept (Z = 5)	0.358	0.393	0.388	0.569
Y^* (Z = 1)	36.660	2.845	1.103	2.730
Y^* (Z = 2)	36.662	2.302	1.433	3.104
Y^* (Z = 3)	36.665	2.322	1.340	2.596
Y^* (Z = 4)	36.674	3.404	1.427	2.735
Y^* (Z = 5)	36.668	2.357	1.405	2.312

Note: Bootstrap standard errors come from 500 bootstrap samples per-list.

Overall, either approach to variance estimation would lead to the same substantive conclusions about the EM coefficients. But each has its benefits and drawbacks. Bootstrapping is straightforward to perform. Additionally, it is agnostic to the choice of distribution for the control count Z_i. So if one wished to try different control models (ordinal logistic, beta-binomial, binomial, etc.) the same bootstrapping approach could be applied. On the other hand, analytic standard errors require one to derive the observed information matrix, which differs between each control distribution (Kuha & Jackson, 2014). But bootstrapping is much slower to implement. The EM algorithm alone can be slow to converge compared to other estimation techniques like Newton-Raphson. Running the model potentially hundreds of times for variance estimation only exacerbates that problem. In this paper, I used parallel processing with the `future` package in R to speed up implementation of the bootstrap. In comparison, deriving analytic standard errors adds only a trivial amount of time to the model.

These conclusions also hold when additional covariates are introduced, but more issues with the bootstrap begin to appear. Table 10 presents the bootstrap ana analytic standard errors from an EM regression model that includes age as an independent variable. Based on the results from section 6 that showed Z_i is independent of Y_i conditional on age, I exclude Y^* as a predictor from these models. Again, in most cases the bootstrap standard errors match the analytic standard errors closely. We see that the list B 65+ coefficient that exhibited signs of perfect separation also has a large bootstrap SE. However, in list A, the bootstrap standard errors in the model for the sensitive item Y_i are much larger than the associated

analytic standard errors. In a small portion of the list A bootstrap samples, the intercept and age coefficients explode (with absolute values around 15), which seems to drive the issue.

Table 10: Comparison of Analytic and Bootstrap Standard Errors for Regression Model

Coefficient	**List A**		**List B**	
	Analytic SE	Bootstrap SE	Analytic SE	Bootstrap SE
Model for sensitive item Y				
Intercept	0.560	1.557	0.510	0.536
Aged 35 to 64	0.760	1.744	0.651	0.988
Aged 65 or over	1.163	6.795	634.905	4.535
Model for control count Z				
Intercept (Z = 1)	0.372	0.406	0.290	0.294
Intercept (Z = 2)	0.416	0.395	0.334	0.283
Intercept (Z = 3)	0.417	0.417	0.345	0.281
Intercept (Z = 4)	0.372	0.453	0.274	0.329
Intercept (Z = 5)	0.419	0.563	0.317	0.304
Age 35–64 (Z = 1)	0.426	0.446	0.344	0.335
Age 35–64 (Z = 2)	0.387	0.450	0.323	0.325
Age 35–64 (Z = 3)	0.428	0.463	0.385	0.321
Age 35–64 (Z = 4)	0.436	0.534	0.313	0.375
Age 35–64 (Z = 5)	0.415	0.622	0.357	0.362
Age 65+ (Z = 1)	0.474	0.461	0.403	0.364
Age 65+ (Z = 2)	0.489	0.446	0.339	0.363
Age 65+ (Z = 3)	0.498	0.471	0.399	0.423
Age 65+ (Z = 4)	0.572	0.522	0.485	0.429
Age 65+ (Z = 5)	0.637	0.723	0.485	0.489

Note: Bootstrap standard errors come from 500 bootstrap samples per-list.

The simple nonparametric bootstrap further breaks down when the EM regression model is misspecified. For example, when Y^* is included in the model above (which is unnecessary as shown in Section 6), the bootstrap standard errors for nearly every coefficient explode in magnitude. Importantly, this does not occur for just the coefficients that exhibited signs of separation (Y^* and 65+), but for every coefficient in the model in both lists A and B. Moreover, the mean of most bootstrap coefficient vectors differed dramatically from the estimated coefficients in the main dataset. As such, using analytic standard errors is likely the preferable approach to variance estimation when applying the EM algorithm to item count data. Beyond speed concerns, they appear to produce more stable estimates under model misspecification. However, the separation issues identified in Section 6 are not an obstacle to accurate item count estimation. Even with perfect separation, we are able to accurately estimate $\hat{\pi}_y$ and how it relates to other variables of interest.

7.2 Control distribution specification

It would have also been possible to carry out EM estimation on this dataset using other distributions for the control questions. Instead of assuming Z_i follows a multinomial distribution, we could instead use a binomial or beta-binomial distribution, for example (Imai, 2011). However, misspecifiying the control count distribution can alter estimates of the sensitive probability $\hat{\pi}_y$ (Kuha & Jackson, 2014). I employ the multinomial distribution in this paper because it is maximally flexible, as recommended by Kuha and Jackson (2014). However, while this approach is safest for avoiding misspecification, the multinomial logistic regression model requires estimating separate coefficients for each category of Z_i. Using a different control distribution would reduce the number of parameters that must be estimated.

To assess my choice of control distribution, I carried out EM estimation on the internet piracy dataset using the `list` package in R (Blair & Imai, 2010). This packages allows for EM item count estimation using either the binomial or beta-binomial distribution for Z_i. I tested both. First, I compared the estimated probability of pirating music, TV shows, or movies in the last year from the multinomial models shown in Section 5 to those from the `list` package. These results are presented in Table 11. For both lists A and B, the beta-binomial model provides very similar estimates $\hat{\pi}_y$ to the multinomial model, when independence between Z_i and Y_i is assumed. For both lists, the point estimates match exactly, but the standard errors are smaller in the beta-binomial models. There are larger differences between the beta-binomial models and the correctly specified multinomial logistic models that assume dependence between Z_i and Y_i (0.20 vs. 0.21 for list A and 0.25 vs. 0.18 for list B).

Table 11: Estimates of π_y from EM Models Assuming Multinomial, Beta-Binomial, and Binomial Distributions for Z_i

List	**Model**	$\hat{\pi}_y$
List A	Multinomial (dependent)	0.205 (0.076)
	Multinomial (independent)	0.199 (0.057)
	Binomial	–
	Beta–binomial	0.203 (0.017)
List B	Multinomial (dependent)	0.180 (0.110)
	Multinomial (independent)	0.250 (0.047)
	Binomial	0.312 (0.037)
	Beta–binomial	0.247 (0.015)

Note: Standard errors in parentheses. "–" indicates that a model failed to converge.

However, it is clear that the binomial distribution is inadequate for these data. In list B, the binomial estimate of $\hat{\pi}_y = 0.31$ differs substantially from the two multinomial models and the beta-binomial model. And in list A, the binomial model from the `list` package failed to converge at all. But at least for the beta-binomial models, one would draw similar

conclusions about the prevalence of internet piracy to those from the multinomial model. However, it is also important to assess whether the conclusions drawn from the regression analysis in Section 6 depend on the choice of control distribution. Therefore, Table 12 presents a comparison of the intercept and age coefficients from my multinomial logistic regression models and regression models from the `list` package that specify the binomial and beta-binomial distribution for the control count Z_i.

Note that these coefficients are for the sensitive model for Y_i, as that is the model of primary interest. In this case, the binomial models for lists A and B failed to converge. That provides even more evidence that the binomial distribution — which is the least flexible control count distribution possible — is not adequate for these item count data. But again, the beta-binomial age coefficients are relatively similar to the multinomial age coefficients. While the magnitudes differ model-to-model, the conclusions one would draw do not: there is a negative relationship between age and the probability of illegally downloading movies, TV shows, or music from the internet.

Table 12: Estimated Regression Coefficients for Y_i from EM Models Assuming Multinomial, Beta-Binomial, and Binomial Distributions for Z_i

Model	**Intercept**	**Aged 35–64**	**Aged 65+**
List A			
EM Dependent	0.37 (0.60)	–2.59 (1.71)	–3.43 (1.44)
EM Independent	–0.31 (0.56)	–0.84 (0.76)	–2.77 (1.16)
Binomial	–	–	–
Beta-binomial	0.35 (0.77)	–1.49 (0.93)	–3.35 (1.26)
List B			
EM Dependent	0.12 (0.74)	–2.11 (1.10)	–16.50 (1.43)
EM Independent	0.35 (0.51)	–1.51 (0.65)	–15.36 (634.90)
Binomial	–	–	–
Beta-binomial	0.60 (0.49)	–1.83 (0.63)	–18.77 (2097.15)

Note: Standard errors in parentheses. "–" indicates that model did not converge.

Overall, it appears that using the beta-binomial distribution to model the control count Z_i would also be adequate for the item count question in the FIDUCIA dataset. That holds for lists A and B. But these results also demonstrate the importance of using a sufficiently flexible control count distribution. The binomial distribution is simply inappropriate for these data and would result in misleading inferences.

8 National Estimates of Internet Piracy

One important limitation of the substantive findings about internet piracy presented in this paper is that the survey was conducted in the UK and Bulgaria. As such, the piracy estimates above only apply to those two countries. But one benefit of a cross-national survey is that we can assess the frequency of illegal piracy in the UK and Bulgaria separately and test whether they differ between the two countries. I begin by simply including a dichotomous variable for respondents' country in the pooled EM model presented in Table 4. I do the same for the separate EM algorithm models for list A and B. Based on the findings from Sections 5 and 6, I fit all three models assuming dependence between Z_i and Y_i. Table 13 presents the regression coefficients for the sensitive model for Y_i from all three models.

These results indicate that residents of the United Kingdom are less likely than those in Bulgaria to engage in internet piracy. That finding holds for both lists A and B and in the pooled model, with the UK coefficient ranging from -2.03 to -2.17. Substantively, that translates to the odds of engaging in internet piracy being between 42% and 57% for those in the UK compared to those in Bulgaria, depending on the dataset. All three coefficients are statistically significant. Additionally, the AICs for each model are smaller than the AICs for the comparable intercept-only models (3458 vs. 3312 for list A, 3371 vs. 3179 for list B, and 6827 vs. 6488 for the pooled model). This indicates that incorporating information about respondents' countries of origin results in a better fitting model for the data. As such, there is clear evidence that the frequency of internet piracy differs between these two countries.

Table 13: Regression Coefficients from the Model for Y_i with Country of Residence Included as a Predictor

Predictor	**List A**	**List B**	**Pooled**
Intercept	-0.854 (0.370)	-0.545 (0.464)	-0.758 (0.259)
Country = UK	-2.026 (0.757)	-2.173 (1.042)	-2.048 (0.465)
AIC	3312.134	3179.728	6488.120
Log-likelihood	-1639.067	-1572.864	-3212.060

Note: Standard errors in parentheses.

Therefore, it is worthwhile to derive separate estimates of illegal piracy for the UK and Bulgaria. Below, I apply the EM algorithm separately to the survey data from the UK and Bulgaria. In each case, I fitted the model separately for list A and B, once assuming dependence between Z_i and Y_i and once assuming independence. The results for the UK are shown in Table 14. For both control item lists, the likelihood ratio test between the model that includes Y^* and the model that excludes it is not statistically significant. Therefore, we fail to reject the null hypothesis of independence. In the UK, a model that assumes Z_i and Y_i are independent is adequate, which is a departure from the EM models used on the full dataset. However, this independence holds for both lists A and B. So again, there is no evidence that using a list of thematically unrelated control questions provides any benefit in this area.

Based on the models that exclude Y^* as a predictor, the list A EM algorithm indicates that the proportion of people in the UK who have engaged in internet piracy in the past year is 0.194. The list B estimate is 0.203. A Wald test for the difference between those estimates indicates that they are not significantly different from one another ($p = 0.933$). As such, there is again no evidence that the related or unrelated control item lists result in significantly different estimates of the sensitive probability $\hat{\pi}_y$.

Table 14: EM Estimates of the Probability of Illegally Downloading Music, TV Shows, or Movies from the Internet in the UK

			Control Count							
List	**Conditional on Y?**	$\hat{\pi}_y$	**0**	**1**	**2**	**3**	**4**	**5**	**AIC**	***p*-value**
List A	Independent	0.194 (0.086)	0.068	0.156	0.221	0.295	0.189	0.072	1811.804	
	($Y = 0$)	0.158 (0.079)	0.064	0.170	0.236	0.283	0.180	0.066	1820.171	0.897
	($Y = 1$)		0.075	0.000	0.199	0.400	0.225	0.101		
List B	Independent	0.203 (0.064)	0.154	0.136	0.177	0.203	0.201	0.129	1743.573	
	($Y = 0$)	0.117 (0.118)	0.158	0.153	0.192	0.199	0.194	0.104	1747.375	0.287
	($Y = 1$)		0.000	0.037	0.058	0.319	0.253	0.333		
Difference	Independent	-0.009								0.933
	Dependent	0.040								0.777

Note: Standard errors in parentheses. The p-values in the rows for the list A and B models come from a likelihood-ratio test between the model assuming independence and the model assuming dependence. The p-values in the difference rows come from Wald tests of the differences between the list A and B sensitive estimates.

Table 15 contains the equivalent regression coefficients for the Bulgaria sample. In these models, the likelihood ratio tests between the independent and dependent models are highly statistically significant. This indicates that in Bulgaria, independence between Z_i and Y_i cannot be assumed. As such, one should rely on the EM estimates from the models that include Y^* as a predictor. For list A, dependent the estimate indicates that the proportion of people engaging in illegal piracy in Bulgaria is 0.284. List B shows the estimate as a larger 0.344. However, based on the Wald test of the difference between these estimates, they are not significantly different from one another ($p = 0.670$). So again, there is no evidence that the choice between a control item list of thematically related on unrelated items alters the sensitive estimate $\hat{\pi}_y$. In Bulgaria, the relationship between Z_i and Y_i is also as expected. Respondents who pirated music, TV shows, or movies in the past year were also more likely to answer "Yes" to more of the control list items than those who had not engaged in internet piracy.

Returning to the country-based differences, we see clear evidence that internet piracy is more common in Bulgaria than in the UK. In list A for example, the proportion of people who have illegally downloaded music, TV shows, or movies in the last year is 0.1 higher in Bulgaria than in the UK (0.284 vs. 0.194). And for list B, the difference is 0.131 (0.334 vs. 0.203). However, based on a Wald test, the difference between these estimates is not statistically significant for either list ($p = 0.445$ for list A and $p = 0.272$ for list B). But because in both countries, there is not a significant difference between the sensitive estimates from lists A and

B, we can pool across lists to get a single estimate $\hat{\pi}_y$ for each country. In these pooled models, the sensitive estimate for the UK is 0.145 and the pooled estimate for Bulgaria is 0.306. The difference between those two estimates (0.161) is marginally statistically significant ($p = 0.06$) based on a Wald test.

Table 15: EM Estimates of the Probability of Illegally Downloading Music, TV Shows, or Movies from the Internet in Bulgaria

			Count							
List	**Conditional on Y?**	$\hat{\pi}_y$	**0**	**1**	**2**	**3**	**4**	**5**	**AIC**	***p*-value**
List A	Independent	0.188 (0.070)	0.223	0.332	0.221	0.121	0.074	0.030	1519.179	0.000
	($Y = 0$)	0.284 (0.085)	0.288	0.396	0.174	0.052	0.083	0.007	1499.086	
	($Y = 1$)		0.000	0.270	0.410	0.231	0.000	0.089		
List B	Independent	0.297 (0.061)	0.468	0.240	0.146	0.073	0.046	0.027	1448.240	0.005
	($Y = 0$)	0.344 (0.112)	0.551	0.270	0.126	0.024	0.029	0.000	1441.269	
	($Y = 1$)		0.264	0.256	0.202	0.157	0.059	0.061		
Difference	Independent	-0.109								0.239
	Dependent	-0.060								0.670

Note: Standard errors in parentheses. The p-values in the rows for the list A and B models come from a likelihood-ratio test between the model assuming independence and the model assuming dependence. The p-values in the difference rows come from Wald tests of the differences between the list A and B sensitive estimates.

9 Conclusions

The item count method provides an elegant solution to the problem of eliciting answers to sensitive survey questions. Compared to techniques like the randomized response method, it deviates less from the standard rhythm of a survey questionnaire but still allows respondents to disguise their answers to sensitive questions. And by treating the results of an item count question as missing categorical data, the expectation maximization (EM) algorithm allows for straightforward point estimation and regression analysis. But the accuracy of the method depends in large part on the design of the list of control items included with the sensitive item.

Improper list design can lead to violations of the basic assumptions of the item count method. Additionally, past work has shown that the control items should be conditionally independent of the sensitive item for efficiency purposes (Kuha & Jackson, 2014). When carrying out item-count estimation, we would like to have the most efficient estimates possible. However, it is challenging to tell whether independence will hold when designing an item count list. Therefore, I employed a novel survey experiment that included two lists of control questions: one that was thematically related to the sensitive item (illegally downloading movies, TV shows, or music) and one that was thematically unrelated. Theoretically, the list of thematically unrelated items should have a better chance of being independent of respondents' answers to the sensitive question.

However, I find that the choice between a thematically related and thematically unrelated list makes little difference for item count estimation. First, the estimated frequency of internet piracy $\hat{\pi}_y$ does not differ significantly between the two control item lists. Second, neither list allows us to assume that the control count Z_i is independent of the sensitive item Y_i. Given the equivalence between the two lists, I pooled the lists together and derived an overall estimate for the frequency of internet piracy in the UK and Bulgaria: 0.197. In other words, approximately 20% of people in the UK and Bulgaria illegally downloaded movies, TV shows, or music in the past year. Importantly, the EM algorithm allowed for straightforward estimation of this probability without observing respondents' answers to the sensitive question directly.

It is also possible to achieve efficiency gains in EM estimation if Z_i is conditionally independent of Y_i given some covariates. In the substantive application considered here, including age as an independent variable is sufficient for conditional independence. Substantively, I find that age is negatively related to internet piracy. In particular, those over age 65 are consistently less likely than those under age 35 to engage in piracy. But importantly, the conditional independence finding holds for both lists. Even here, there is no noticeable benefit from using a list of thematically unrelated control items instead of a list of related items.

There findings have important implications for the design of item count method questions. Based on the substantive application presented in this paper, there are no distinguishable benefits from using a list of control items that are thematically unrelated to the sensitive question. Both lists produce substantive similar estimates of the sensitive item of interest, and responses to both lists are similarly dependent on responses to the sensitive item. Therefore, when designing an item count question, one should default to a list of control questions that

are thematically related to the sensitive question. That is because the overall item count list will look more coherent to survey respondents if the items are somewhat related. In turn, that will reduce the possible "weirdness factor" that comes from being asked to select how many items in a list apply to you without naming the specific items. Based my analysis of the FIDUCIA experiment, the primary potential downside from using thematically related questions (introducing correlation between Z_i and Y_i) is not a major concern.

The statistical tests that underpin these findings rely in part on accurate variance estimation. Here, I relied on the observed information matrix as laid out by Kuha and Jackson (2014). I also compared the standard errors derived from the information matrix to those from bootstrapping. In most cases, the two were substantively similar. However, the fact the bootstrapping can be prohibitively slow and potentially unstable in poorly specified models led me to conclude that the analytic standard errors were preferable. My results are similarly robust to the choice of distribution for the control count Z_i. Using a beta-binomial distribution to model the count count would lead to broadly similar conclusions about both the frequency of internet piracy and the relationship between age and piracy. However, there is clear evidence that the overly-restrictive binomial distribution is inadequate for these data.

Finally, there is evidence that the probability of pirating music, movies, or TV shows differs between the UK and Bulgaria. In the UK, the probability of engaging in internet piracy is 0.145, compared to 0.306 in Bulgaria. These results point to one obvious limitation of the substantive conclusions presented in this paper: location. To learn how prevalent internet piracy is in other countries, new item count surveys should be conducted in those countries. That is especially important given the evidence present here of non-trivial cross-national differences in the frequency of piracy.

There are other limitations to the findings presented in this paper. First, I employed a single item count question to compare the efficacy of thematically related and unrelated control item lists. It is possible that for a different set of sensitive and control questions asked to a different population, the choice of an unrelated list could provide a material benefit. As such, it would be beneficial to replicate the survey experiment presented in this paper using a different sensitive question and different lists of control questions to confirm the equivalence demonstrated here. However, my findings still provide strong evidence that the choice of a related or unrelated list will not substantively alter the results of an item count question.

Second, my findings only show that a list of thematically unrelated questions is no better at achieving conditional independence between Z_i and Y_i than a list of related questions. It is entirely possible that another aspect of list design could fix this issue. Future work is therefore needed on which aspects of question design — other than thematic similarity — can be altered to achieve independence between Z_i and Y_i. For example, one could conduct an initial item count survey and perform regression analysis to identify particular behaviors that are uncorrelated with the sensitive question. Subsequently, a second survey could be conducted using those behaviors found to be unrelated to the sensitive question as control items.

Finally, the point estimates presented in this paper have an important limitation: they were estimated without the use of survey weights. In this case, the UK dataset does include

a survey weight but the Bulgaria data does not. In general, item count estimation (and estimation on survey datasets in general) should be performed incorporating all survey design elements, including weights, clustering, and stratification. Incorporating weights is trivial for moment-based item count estimation, but slightly more involved when using the EM algorithm.

As in standard estimation problems with complex survey data, one would replace the maximum likelihood estimation carried out in the EM algorithm with maximum pseudolikelihood estimation (Lumley, 2024). The standard log-likelihood function, would be replaced with a weighted log-likelihood where each respondent's contribution to the likelihood is multiplied by their survey weight. That weighted log-likelihood would then be maximized as normal. Importantly, variance estimation could no longer be carried out using the usual information matrix as done in this paper. Future work should add this functionality to packages for EM estimation for item count questions. In appendix 3, I present weighted estimates of internet piracy using a moment-based estimator. My findings are unchanged by introducing the survey weight.

Despite these limitations, the results presented in this paper provide clear evidence that lists of control question that are thematically unrelated to the sensitive item provide no noticeable benefit over thematically related lists. Both kinds of lists result in similar item count estimates that are similarly efficient. Survey researchers should therefore feel confident writing item count lists that fit together coherently and are less likely to confuse survey respondents.

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Appendix

Appendix 1: Acknowledgment: The use of generative AI tools

I acknowledge the use of ChatGPT (OpenAI, https://chatgpt.com/) to assist me in the production of this dissertation. I primarily used ChatGPT 4o (OpenAI, https://openai.com/index/hello-gpt-4o/) and ChatGPT o3 (OpenAI, https://openai.com/index/introducing-o3-and-o4-mini/ However, I switched to using ChatGpt 5.0 (OpenAI, https://openai.com/index/introducing-gpt-5/) when that model was released in early August, 2025.

I primarily used ChatGPT to diagnose and solve bugs in the R code used in this dissertation (R Core Team, 2012). Generally, this involved providing the model with a problematic section of code and a description of the issue (or the errors generated by R), and asking it to identify potential problems and/or recommend solutions to those problems. Most of this troubleshooting was carried out on the EM estimation function I wrote in R for this paper (lines 150 to 669 in the R script that can be downloaded here).

I also used ChatGPT to produce the BibTeX citations used in this dissertation. I provided the model with information about each paper I referenced (either in the form of a PDF or a URL) and asked it to produce a BibTeX citation for that paper. I then pasted that BibTeX into the .bib file associated with the references section of this dissertation. Additionally, I used ChatGPT to check this dissertation for typos. I did so by providing it with my final draft and asking it to point out places where I had misspelled words.

Finally, I used ChatGPT to assist in creation and formatting of the LaTeX tables included in this paper. This process took two forms. First, when I wished to edit the look of a table and did not know how to do so in LaTeX, I provided the table to ChatGPT and asked it to make the necessary change to the LaTeX code. For example, I used it to add footnotes to my tables. Second, when I made similar tables with different statistical output, I used ChatGPT to update my table templates with new output. For example, Tables 6 and 8 are structurally similar. To create Table 8, I provided the model with Table 6 and the output for Table 8 and had it put the new output into the same LaTeX format.

Appendix 2: The FIDUCIA survey

In Section 4 I described the public opinion survey considered in this paper. Here, I provide additional information about the FIDUCIA survey. The UK wave of the survey was conducted by Opinion Research Services. Their survey of 1,004 British adults had a response rate of 11.7% and was weighted by ethnicity, age, gender, and working status, interacted with five regions: North, Wales and South West, London, South East, and East and Midlands. The Bulgaria wave was conducted by Alpha Research and resulted in 1,024 complete interviews. That survey had a response rate of 70%. It did not include a survey weight.

Each survey provider included a summary table of the descriptive statistics for their final samples. Below, Figure 3 shows that table for the UK sample. It includes both the weighted and unweighted figures, and how they compare to the population of interest.

Characteristic	Unweighted Count	Unweighted Valid %	Weighted Valid %	Resident Population %
BY AGE				
18 to 24	134	13.3%	11.9%	11.9%
25 to 34	129	12.8%	17.1%	17.1%
35 to 54	291	29.0%	35.2%	35.2%
55 or over	450	44.8%	35.8%	35.8%
Total valid responses	**1,004**	**100%**	**100%**	**100.0%**
BY GENDER				
Male	469	46.7%	47.2%	48.6%
Female	535	53.3%	52.8%	51.4%
Total valid responses	**1,004**	**100.0%**	**100.0%**	**100.0%**
BY WORKING STATUS				
Working	584	58.2%	60.4%	57.9%
Retired	276	27.5%	20.2%	22.0%
Otherwise not working	144	14.3%	19.3%	20.1%
Total valid responses	**1,004**	**100.0%**	**100.0%**	**100.0%**
BY ETHNICITY				
White	930	93.0%	87.8%	87.8%
Non White	70	7.0%	12.2%	12.2%
Total valid responses	**1,000**	**100.0%**	**100.0%**	**100.0%**
Unknown	*4*	-	-	-
BY REGION				
North	268	26.7%	26.7%	26.7%
East and Midlands	285	28.4%	28.4%	28.4%
Wales and South West	152	15.1%	15.1%	15.1%
London	145	14.4%	14.4%	14.4%
South East	154	15.3%	15.4%	15.4%
Total valid responses	**1,004**	**100.0%**	**100.0%**	**100.0%**

Figure 3: Descriptive Statistics for the UK FIDUCIA Wave

Similarly, Figure 4 shows the descriptive statistics table for the Bulgaria sample. Note that the racial and regional groups shown in each table differ, which is unavoidable in a cross-national survey.

		Count	Col %
Age:	16-17	15	1,5%
	18-30	174	17,0%
	31-40	178	17,4%
	41-50	160	15,6%
	51-60	179	17,5%
	61 +	314	30,7%
	N/A	4	0,3%
Total		1024	100,0%

		Count	Col %
Gender:	Male	495	48,3%
	Female	529	51,7%
Total		1024	100,0%

		Count	Col %
Ethnic group?	Bulgarian	910	88,9%
	Turkish	70	6,8%
	Roma	31	3,0%
	Other	3	,3%
	N/A	10	1,0%
Total		1024	100,0%

		Count	Col %
Region	Blagoevgrad	44	4,3%
	Burgas	50	4,9%
	Varna	60	5,9%
	V. Tarnovo	32	3,1%
	Vidin	20	2,0%
	Vratza	22	2,1%
	Gabrovo	20	2,0%
	Dobrich	32	3,1%
	Kardzhali	20	2,0%
	Kyustendil	20	2,0%
	Lovech	20	2,0%
	Montana	30	2,9%
	Pazardzhik	42	4,1%
	Pernik	20	2,0%
	Pleven	32	3,1%
	Plovdiv	88	8,6%
	Razgrad	20	2,0%
	Ruse	32	3,1%
	Silistra	20	2,0%
	Sliven	30	2,9%
	Smolyan	22	2,1%
	Sofia	200	19,5%
	Stara Zagora	52	5,1%
	Targovishte	22	2,1%
	Haskovo	32	3,1%
	Shumen	22	2,1%
	Yambol	20	2,0%
Total		1024	100,0%

		Count	Col %
Type of settlement:	Capital	200	19,5%
	Regional city	342	33,5%
	Small town	200	19,5%
	Village	282	27,5%
Total		1024	100,0%

Figure 4: Descriptive Statistics for the Bulgaria FIDUCIA Wave

Appendix 3: Weighted Item Count Results

As mentioned in Section 9, an ideal analysis of the data presented in this paper would incorporate survey weights. In this section, I present moment-based estimates of the item

count question about internet piracy, incorporating the weights provided by the FIDUCIA survey. Note that only the UK wave provided survey weights. As such, the weight for each Bulgarian respondent is set to 1 for this analysis. To estimate the weighted $\tilde{\pi}_{yw}$, I simply take the difference in weighted means between the treatment group and control group:

$$\tilde{\pi}_{yw} = \frac{1}{\sum_{i=1}^{n_1} w_{i(1)}} \sum_{i=1}^{n} t_i s_i w_{i(1)} - \frac{1}{\sum_{i=1}^{n_0} w_{i(0)}} \sum_{i=1}^{n} (1 - t_i) s_i w_{i(0)} \tag{18}$$

where $w_{i(1)}$ is the vector of survey weights for respondents in the treatment group and $w_{i(0)}$ is the vector of weights for those in the control group. The control count probabilities are calculated using Equations 4 and 5 (Kuha & Jackson, 2014). However, $\hat{p}_s$ is no longer the observed proportion of each response to s_i. Instead, I calculate the weighted proportions using the survey weight w_i. Table 16 shows the weighted moment-based estimates separately for lists A and B.

We see that the weighted estimates differ slightly from the unweighted moment-based estimates. The different is larger for list A (0.186 vs. 0.146) than it is for list B (0.182 vs. 0.180). In both cases, the weighted estimate is higher than the unweighted estimate. But again, we see that the difference in the sensitive probability estimates $\tilde{\pi}_{yw}$ between lists A and B is small: 0.0038. The control count estimates are also similar in the weighted and unweighted models. In both cases, list A has a negative probability. Additionally, we again see that respondents who answered "Yes" to the sensitive question are more likely to answer "Yes" to all five control questions and less likely to answer "Yes" to zero control questions than are respondents to answered "No" to the sensitive question.

Table 16: Weighted Moment-Based Estimates of the Probability of Illegally Downloading Music, TV Shows, or Movies from the Internet

			Control Count					
List	**Conditional on Y?**	$\hat{\pi}_y$	**0**	**1**	**2**	**3**	**4**	**5**
List A	$(Y = 0)$	0.186	0.167	0.283	0.215	0.171	0.126	0.038
	$(Y = 1)$		-0.140	0.162	0.362	0.354	0.148	0.113
List B	$(Y = 0)$	0.182	0.310	0.201	0.176	0.141	0.118	0.055
	$(Y = 1)$		0.185	0.172	0.124	0.210	0.124	0.186
Difference		0.0038						

Clearly, these data should be analyzed with survey weights. However, in this case, the inclusion of the weight does not appear to alter the substantive conclusions in this paper. There is still little difference in the sensitive probability estimates between the related and unrelated lists. And around 20% of the population are estimated to have engaged in internet piracy within the past year, a similar proportion to the unweighted analysis.